

Student Course Enrollment & Attendance Tracker

The **Student Course Enrollment & Attendance Tracker** is a Salesforce CRM-based solution designed for academic institutions to digitize and streamline enrollment and attendance processes. It helps administrators manage student records, course enrollments, and attendance efficiently while providing teachers with tools to mark and monitor participation. Key features include automated alerts for low attendance, real-time reporting, and detailed course-wise statistics. This CRM addresses the business need for reduced administrative workload, minimized manual errors, and improved communication between teachers, students, and administrators.

The primary objective of building this CRM is to simplify and automate enrollment and attendance management for educational institutions. It ensures accurate student-course data, enables quick attendance tracking, and delivers proactive alerts to students and guardians. By offering comprehensive reports on attendance and enrollment trends, the system enhances decision-making and supports compliance requirements. Ultimately, the CRM improves academic monitoring, increases student engagement, and boosts institutional efficiency—delivering strong business value through better operational management and performance tracking.

Phase 1 Student Course Enrollment & Attendance Tracker

1. Requirement Gathering

The project focuses on solving challenges in educational institutions regarding student enrollment and attendance tracking. The key requirements gathered are:

- Efficient maintenance of student and course records.
- Seamless enrollment of students into their respective courses.
- Attendance marking for each class session.
- Automatic alerts for students with low attendance.
- Generation of reports including attendance percentages and enrollment statistics.

The system aims to reduce manual work, minimize errors, and improve communication between students, teachers, and administrators.

2. Stakeholder Analysis

The primary stakeholders are:

- **Students** – use the system for enrollment, attendance tracking, and receiving notifications.
- **Teachers** – manage attendance marking and participation tracking.
- **Academic Administrators** – monitor attendance trends, compliance, and reporting.

Each stakeholder benefits from streamlined processes and timely information that supports better academic engagement and performance.

3. Business Process Mapping

The business processes involved include:

- Recording and updating student and course information.
- Enrolling students into courses based on academic schedules.
- Tracking attendance for each class session.
- Sending automated alerts when attendance falls below thresholds.
- Generating attendance reports for teachers and administrators to analyze trends and take corrective actions.

4. Industry-specific Use Case Analysis

The project serves the education sector (EdTech / Academic Institutions), especially in schools, colleges, and training centers. The following use cases were identified:

- Automating attendance tracking to reduce manual intervention.

- Alerting students or guardians to improve attendance and academic performance.
- Providing administrators with actionable insights for academic planning.
- Facilitating accurate record-keeping for compliance and reporting.

5. AppExchange Exploration

Salesforce CRM is leveraged as the underlying platform for the application, enabling:

- Centralized data management for students and courses.
- Automated workflows and notifications.
- Report generation capabilities for academic administrators.
- Role-based access for teachers and students.

This ensures scalability, ease of maintenance, and enhanced user experience.

Phase 2: Org Setup & Configuration

Project: **STUDENT COURSE ENROLLMENT & ATTENDANCE TRACKER**

Institute: Prof. Ram Meghe Institute of Technology & Research

1. Salesforce Edition

- Developer Org used for implementation.

2. Company Profile Setup

- Organization Name: Prof. Ram Meghe Institute of Technology & Research

- Default Locale: English (India)

- Default Currency: INR

- Default Time Zone: (GMT+05:30) India Standard Time

Process Flow: Setup → Company Information → Edit Company Profile

SETUP

Company Information

The organization's profile is below.

User Licenses (10)

 |

Permission Set Licenses (10)

 |

Feature Licenses (1)

 |

Usage-based Entitlements (10)

Organization Detail

Edit

Organization Name	Prof. Ram Meghe Institute of Technology & Research	Phone	
Primary Contact	OrgFarm EPIC	Fax	
Division		Default Locale	English (India)
Address	India	Default Language	English
Fiscal Year Starts In	Custom Fiscal Year	Default Time Zone	(GMT+05:30) India Standard Time (Asia/Kolkata)
Activate Multiple Currencies	☐	Currency Locale	Hindi (India) - INR
Enable Data Translation	☐	Used Data Space	449 KB (9%) View
Newsletter	☒	Used File Space	17 KB (0%) View
Admin Newsletter	☒	API Requests, Last 24 Hours	2 (15,000 max)
Hide Notices About System Maintenance	☐	Streaming API Events, Last 24 Hours	0 (10,000 max)
Hide Notices About System Downtime	☐	Restricted Logins, Current Month	0 (0 max)
Locale Formats	ICU	Salesforce.com Organization ID	00DgK000007Zu1j
		Organization Edition	Developer Edition
		Instance	CAN96

Created By

OrgFarm EPIC, 7/17/2025, 9:17 AM

Edit

Modified By

Harshal Alaspure, 9/15/2025, 5:46 AM

3. Business Hours & Holidays

- Business Hours: "School Hours" (All days, 09:00 AM – 05:00 PM)

- Holidays: Configurable (none added yet).

■ Process Flow: Setup → Company Settings → Business Hours → Define & Assign

The screenshot shows the 'Business Hours' setup page. At the top, there's a 'SETUP' header with a building icon and the title 'Business Hours'. Below this, a note states: 'If you enter blank business hours for a day, that means your organization does not operate on that day.' The main section is titled 'Business Hours Edit' and includes 'Save' and 'Cancel' buttons. It is divided into three steps:
1. **Step 1. Business Hours Name**: Contains a text field for 'Business Hours Name' with the value 'School Hours', a checkbox for 'Active' (unchecked), and a checkbox for 'Use these business hours as the default' (unchecked).
2. **Step 2. Time Zone**: Features a dropdown menu for 'Time Zone' set to '(GMT+05:30) India Standard Time (Asia/Kolkata)'.
3. **Step 3. Business Hours**: A table showing hours for each day of the week. Each row has a time input field (e.g., '9:00 AM') and a '24 hours' checkbox (all unchecked).
At the bottom of the form are 'Save' and 'Cancel' buttons.

4. Fiscal Year Settings

- Fiscal Year: Custom, 01-Apr-2025 to 31-Mar-2026

■ Process Flow: Setup → Company Settings → Fiscal Year → Custom Fiscal Year

The screenshot shows the 'Custom Fiscal Year Definition' page. The top header includes a 'SETUP' icon and the title 'Custom Fiscal Year Definition'. A 'Help for this Page' link is visible. A descriptive note explains that the page allows defining a custom fiscal year record based on user choices. The main form area is titled 'Fiscal Year Information' and includes 'Preview', 'Save', and 'Cancel' buttons. It contains the following fields:
- 'Fiscal Year Start Date': A date picker set to '4/1/2025'.
- 'Fiscal Year Name': A text field containing '2026'.
- 'Description': A large text area for additional details.
- 'Week Start Day': A dropdown menu set to 'Tuesday'.
Below these fields is a section titled 'Reset Fiscal Year Structure'. At the bottom of the form are 'Preview', 'Save', and 'Cancel' buttons.

5. User Setup & Licenses

- Teacher User: License = Salesforce, Profile = Teacher_Profile, Role = Teacher
- Student User: License = Salesforce, Profile = Student_Profile, Role = Student
- Guardian User: License = Salesforce, Profile = Guardian_Profile, Role = Guardian

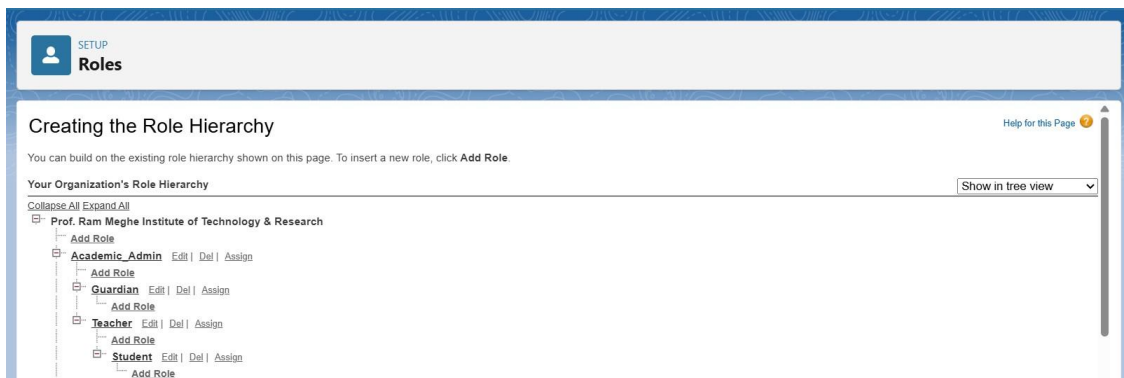
Process Flow: Setup → Users → New User → Assign License & Profile

The screenshot shows the 'New User' setup page in Salesforce. The page is titled 'New User' and has a 'User Edit' section with 'Save', 'Save & New', and 'Cancel' buttons. The 'General Information' tab is active, showing fields for First Name (Anita), Last Name (Deshmukh), Alias (adesh), Email, Username, Nickname, Title, Company, Department, and Division. On the right, there are dropdown menus for Role (Teacher), User License (Salesforce), and Profile (None). There are also checkboxes for Active, Marketing User, Offline User, Knowledge User, Flow User, Service Cloud User, Site.com Contributor User, Site.com Publisher User, and WDC User. At the bottom, there are dropdowns for Data.com User Type (None) and Data.com Monthly Addition Limit (Default Limit (300)).

6. Roles Hierarchy

- Academic_Admin → Teacher → Student
- Guardian under Admin

Process Flow: Setup → Roles → Define Hierarchy → Assign Users



7. Profiles

- Teacher_Profile (cloned and customized from Standard User)
- Student_Profile (cloned and customized from Standard User)
- Guardian_Profile (cloned and customized from Standard User)

■ Process Flow: Setup → Profiles → Clone Standard Profile → Customize Permissions

The screenshot shows the 'Clone Profile' page in the Salesforce Setup interface. The page title is 'Clone Profile' with a 'Help for this Page' link. Below the title, it says 'Enter the name of the new profile.' A message box states 'You must select an existing profile to clone from.' Below this, there are three fields: 'Existing Profile' with a dropdown menu showing 'Standard User', 'User License' with a dropdown menu showing 'Salesforce', and 'Profile Name' with a text input field containing 'Teacher_Profile'. At the bottom, there are 'Save' and 'Cancel' buttons.

8. Permission Sets

- PS_Grade_Manager → grants extra edit rights on Grade_c, assigned to Teacher user (Anita).
- PS_Attendance_Manager → grants rights to manage Attendance_c, assigned to Teacher user (Anita).

■ Process Flow: Setup → Permission Sets → Create → Assign to User

The screenshot shows the 'Permission Sets' page in the Salesforce Setup interface. The page title is 'Permission Sets' with a 'Help for this Page' link. Below the title, it says 'On this page you can create, view, and manage permission sets.' There is a filter dropdown set to 'All Permission Sets' and links for 'Edit', 'Delete', and 'Create New View'. Below this is a table with columns: 'Action', 'Permission Set Name', 'Description', and 'License'. The table contains two rows of data.

Action	Permission Set Name	Description	License
Del Clone	PS_Attendance_Manager		
Del Clone	PS_Grade_Manager	Allows teachers to create/edit/delete Grade_c records	

9. Organization-Wide Defaults (OWD)

- Student__c = Private
- Guardian__c = Private
- Attendance__c = Private
- Grade__c = Private
- Course__c = Public Read Only

■ Process Flow: Setup → Sharing Settings → Define OWD

The screenshot shows the 'Sharing Settings' page in Salesforce Setup. The page has a left sidebar with a list of objects and their sharing settings. The main area displays a table with columns for the object name, the sharing model (Public Read/Write, Public Read Only, Private), and a checkbox for 'Control Access'. The 'Course' object is highlighted in blue. Below the table, there are 'Other Settings' including 'Standard Report Visibility' (checked), 'Manual User Record Sharing' (unchecked), 'Manager Groups' (unchecked), 'Secure guest user record access' (checked), and 'Require permission to view record names in lookup fields' (unchecked). At the bottom, there are 'Save' and 'Cancel' buttons.

Object	Sharing Model	Control Access
Streaming Channel	Public Read/Write	☒
Tableau Host Mapping	Public Read Only	☒
User Presence	Public Read Only	☒
Waitlist	Private	☒
Web Cart Document	Private	☒
Work Order	Private	☒
Work Plan	Private	☒
Work Plan Template	Private	☒
Work Step Template	Private	☒
Work Type	Private	☒
Work Type Group	Public Read/Write	☒
Attendance	Private	☒
Course	Public Read Only	☒
Grade	Private	☒
Guardian	Private	☒
Student	Private	☒

Other Settings

Standard Report Visibility ☒ ⓘ

Manual User Record Sharing ☐ ⓘ

Manager Groups ☐ ⓘ

Secure guest user record access ☒ ⓘ

Require permission to view record names in lookup fields ☐ ⓘ

Save Cancel

10. Sharing Rules

- Grades (Grade__c): Shared with Guardian role.
- Attendance (Attendance__c): Shared with Guardian role if needed.
- Assignments: Shared with Teacher role.

■ Process Flow: Setup → Sharing Settings → New Sharing Rule → Define Access

SETUP

Sharing Settings

Setup

Attendance Sharing Rule

Help for this Page

Use sharing rules to make automatic exceptions to your organization-wide sharing settings for defined sets of users.

Note: "Roles and subordinates" includes all users in a role, and the roles below that role.

You can use sharing rules only to grant wider access to data, not to restrict access.

Label

Share_Attendance_to_Gua

Rule Name

Share_Attendance_to_Gua

Description

Attendance: owned by members of

Role: Student

Share with

Role: Guardian

Access Level

Read Only

Created By

Harshal Alaspure, 9/15/2025, 6:37 AM

Modified By

Harshal Alaspure, 9/15/2025, 6:37 AM

Save

Cancel

☒ Outcomes of Phase 2:

- Developer Org successfully configured.
- Company academic information standardized.
- Academic roles, profiles, and permission sets created and assigned.
- OWD and Sharing Rules enforced for record-level security.
- Teachers and Guardians now have proper controlled access to Assignments, Attendance, and Grades.
- Phase 2 deliverables completed. Ready to proceed to Phase 3: Data Modeling & Relationships.

Phase 3: Data Modeling & Relationships

1. Standard & Custom Objects

➡ **Process Flow:** *Setup → Object Manager → New Custom Object → Add Fields. All objects configured in a similar manner, with respective fields as listed above.*

Custom objects created for the Assignment & Exam Management System:

Student__c

- Student ID (Text, External ID, Unique)
- First Name, Last Name, Email, Phone
- Guardian (Lookup), Guardian Name, Guardian Contact
- Owner, Created By, Modified By

The screenshot shows the 'New Custom Object' setup page in Salesforce. The breadcrumb navigation at the top is 'Setup > Home > Object Manager > New Custom Object'. A yellow banner at the top states: 'Permissions for this object are disabled for all profiles by default. You can enable object permissions in permission sets or by editing custom profiles. [Tell me more!](#) [Don't show this message again](#)'. Below this is the 'Custom Object Definition Edit' section with 'Save', 'Save & New', and 'Cancel' buttons. The 'Custom Object Information' section includes: 'The singular and plural labels are used in tabs, page layouts, and reports.' with 'Label' set to 'Student' (Example: Account) and 'Plural Label' set to 'Students' (Example: Accounts); 'Starts with vowel sound' is unchecked; 'The Object Name is used when referencing the object via the API.' with 'Object Name' set to 'Student' (Example: Account); a 'Description' text area; 'Context Sensitive Help Setting' with 'Open the standard Salesforce.com Help & Training window' selected; and 'Context Name' set to 'None'. At the bottom, the 'Enter Record Name Label and Format' section shows 'Record Name' set to 'Student Name' (Example: Account Name).

Guardian__c

- Guardian ID (Auto Number, External ID, Unique)
- First Name, Last Name, Email (Unique), Phone
- Relationship to Student (Picklist)
- Owner, Created By, Modified By

SETUP
New Custom Object

Permissions for this object are disabled for all profiles by default. You can enable object permissions in permission sets or by editing custom profiles. [Tell me more!](#) [Don't show this message again](#)

Custom Object Definition Edit Save Save & New Cancel

Custom Object Information Required Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label **Example: Account**

Plural Label **Example: Accounts**

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name **Example: Account**

Description

Context Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page

Content Name

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name **Example: Account Name**

Course__c

- Course Code (Unique), Name (Indexed)
- Credits, Instructor (Lookup), Schedule Info, Title
- Owner, Created By, Modified By

SETUP > OBJECT MANAGER
Course

Details

Fields & Relationships 9 Items, Sorted by Field Label Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Course Code	Course_Code__c	Text(20) (Unique Case Insensitive)		✓
Course Name	Name	Text(80)		✓
Created By	CreatedById	Lookup(User)		
Credits	Credits__c	Number(2, 0)		
Instructor	Instructor__c	Lookup(User)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User, Group)		✓
Schedule Info	Schedule_Info__c	Long Text Area(32768)		
Title	Title__c	Text(255)		

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers

Enrollment__c (Junction Object)

- Enrollment Number (Auto Number)
- Course (Master-Detail), Student (Master-Detail)
- Enrollment Date, Status (Picklist)
- Created By, Modified By

Enrollment

Details

Fields & Relationships

7 Items, Sorted by Field Label

Quick Find

FIELD LABEL	* FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Course	Course__c	Master-Detail(Course)		✓
Created By	CreatedById	Lookup(User)		
Enrollment Date	Enrollment_Date__c	Date		
Enrollment Number	Name	Auto Number		✓
Last Modified By	LastModifiedById	Lookup(User)		
Status	Status__c	Picklist		
Student	Student__c	Master-Detail(Student)		✓

Attendance_c

- Attendance Record (Auto Number), Date
- Enrollment (Lookup), Marked By (Lookup), Status, Remarks
- Owner, Created By, Modified By

SETUP > OBJECT MANAGER
Attendance

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

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Record Types

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Object Access

Triggers

Fields & Relationships

9 Items, Sorted by Field Label

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Attendance Record	Name	Auto Number		✓
Created By	CreatedById	Lookup(User)		
Date	Date__c	Date		
Enrollment	Enrollment__c	Lookup(Enrollment)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Marked By	Marked_By__c	Lookup(User)		✓
Owner	OwnerId	Lookup(User.Group)		✓
Remarks	Remarks__c	Text Area(255)		
Status	Status__c	Picklist		

Grade_c

- Grade Number (Auto Number)
- Enrollment (Lookup), Student (Lookup), Subject (Lookup Course)
- Teacher (Lookup), Marks, Grade Date
- Owner, Created By, Modified By

SETUP > OBJECT MANAGER				
Grade				
Details	Fields & Relationships			
Fields & Relationships	10 Items, Sorted by Field Label			
Page Layouts	Q Quick Find New Deleted Fields Field Dependencies Set History Tracking			
Lightning Record Pages	FIELD LABEL	FIELD NAME	DATA TYPE	INDEXED
Buttons, Links, and Actions	Created By	CreatedById	Lookup(User)	
Compact Layouts	Enrollment	Enrollment__c	Lookup(Enrollment)	✓
Field Sets	Grade Number	Name	Auto Number	✓
Object Limits	Grade Date	Grade_Date__c	Date	
Record Types	Last Modified By	LastModifiedById	Lookup(User)	
Related Lookup Filters	Marks	Marks__c	Number(3, 0)	
Restriction Rules	Owner	OwnerId	Lookup(User,Group)	✓
Scoping Rules	Student	Student__c	Lookup(Student)	✓
Object Access	Subject	Subject__c	Lookup(Course)	✓
Triggers	Teacher	Teacher__c	Lookup(Teacher)	✓

2. Page Layouts

Configured for Student__c as an example:

Process Flow: Setup → Object Manager → Student → Page Layouts → Student Layout → Drag fields into sections → Save.

SETUP > OBJECT MANAGER

Student

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Page Layouts

1 Items, Sorted by Page Layout Name

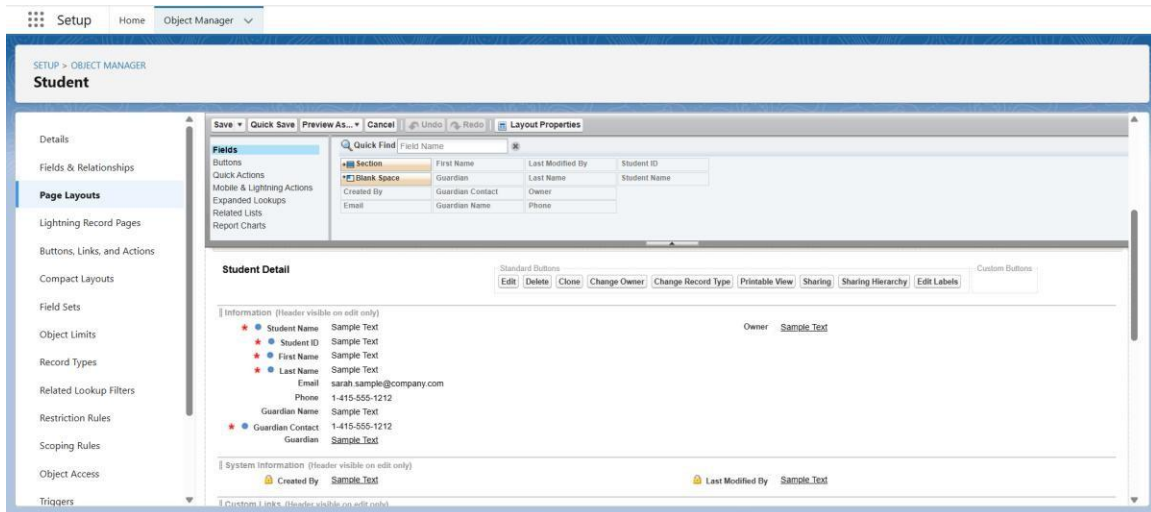
Q Quick Find

New

Page Layout Assignment

PAGE LAYOUT NAME	CREATED BY	MODIFIED BY
Student Layout	Harshal Alaspure, 9/14/2025, 7:52 PM	Harshal Alaspure, 9/16/2025, 10:13 AM

Fields grouped into Student Information and Guardian Info sections, with related lists (Guardians, Enrollments, Grades).



Other objects (Guardian, Course, Enrollment, Attendance, Grade) had layouts configured similarly with their relevant fields.

6. Lookup vs Master-Detail vs Hierarchical Relationships

Lookup Relationship: Attendance linked to Enrollment via Lookup.

Master-Detail Relationship: Enrollment links to Student and Course.

Hierarchical Relationship: User roles such as Teacher under Academic_Admin.

7. Junction Objects

Enrollment_c functions as a junction object connecting Student_c and Course_c. It enables many-to-many relationships between students and courses.

The screenshot shows the 'Enrollment' object fields and relationships configuration in Salesforce Setup. The left sidebar lists various configuration options, with 'Fields & Relationships' selected. The main area displays a table of fields and relationships for the 'Enrollment' object. The table has five columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The table contains seven rows of data, including fields like Course, Created By, Enrollment Date, Enrollment Number, Last Modified By, Status, and Student.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Course	Course_c	Master-Detail(Course)		✓
Created By	CreatedById	Lookup(User)		
Enrollment Date	Enrollment_Date_c	Date		
Enrollment Number	Name	Auto Number		✓
Last Modified By	LastModifiedById	Lookup(User)		
Status	Status_c	Picklist		
Student	Student_c	Master-Detail(Student)		✓

Phase 4 Completion Report – Process Automation (Admin)

1. Validation Rules

Validation rules ensure data integrity by enforcing constraints on field values. The following validation rules were created across different objects:

Steps to Create Validation Rule (applies to all rules):

1. Go to Setup → Object Manager → Select the required Object (e.g., Student).
2. Navigate to Validation Rules → New Rule.
3. Enter Rule Name and Description.
4. Define the Formula (logic specific to the rule).
5. Enter the Error Message and select Error Location.
6. Save & Activate.

Repeating repeat the steps for each rule as listed above.

- Guardian_Required (Student) – Ensure Guardian is linked to every Student.

The screenshot shows the 'Validation Rule Edit' interface in Salesforce. The breadcrumb trail at the top reads 'SETUP > OBJECT MANAGER > Student'. The left sidebar contains a navigation menu with the following items: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, and Triggers. The main content area is titled 'Validation Rule Edit' and includes a 'Save' button, a 'Save & New' button, and a 'Cancel' button. The 'Rule Name' field is populated with 'Guardian_Required'. The 'Active' checkbox is checked. The 'Description' field is empty. Below the description is the 'Error Condition Formula' section. It includes an example: 'Example: [Discount_Percent__c] > 0.30' with a link to 'More Examples...'. Below the example, it states: 'Display an error if Discount is more than 30%. If this formula expression is true, display the text defined in the Error Message area.' The formula field contains the text: 'ISBLANK([Guardian__c])'. To the right of the formula field is a 'Functions' dropdown menu showing a list of functions: ABS, ACOS, ADDMONTHS, AND, ASCII, and ASIN. Below the list is an 'Insert Selected Function' button. At the bottom of the formula field is a 'Check Syntax' button. A 'Quick Tips' sidebar on the right shows a link to 'Operators & Functions'. A red error message at the top of the main content area reads: 'The save will be aborted and the error message will be displayed. The user can correct the error and try again.'

SETUP > OBJECT MANAGER

Student

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
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Record Types
Related Lookup Filters
Restriction Rules
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If this formula expression is **true**, display the text defined in the Error Message area

Insert Field
Insert Operator

ISBLANK(Guardian__c)

Check Syntax

ABS
ACOS
ADDMONTHS
AND
ASCII
ASIN

Insert Selected Function
ABS(number)
Returns the absolute value of a number, a number without its sign

Help on this function

Error Message

Example: Discount percent cannot exceed 30%

This message will appear when Error Condition formula is **true**

Error Message: Every student must be associated with a Guardian

This error message can either appear at the top of the page or below a specific field on the page

Error Location: ☐ Top of Page ☒ Field Guardian

Save Save & New Cancel

- Validate_Email_Format (Student) – Ensure Student email is in correct format.

SETUP > OBJECT MANAGER

Student

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Student Validation Rule

[Back to Student](#)

Validation Rule Detail			
Rule Name	Validate_Email_Format	Active	☒
Error Condition Formula	NOT(ISBLANK(Email__c)) && NOT(CONTAINS(Email__c, "@"))		
Error Message	Please enter a valid email address.	Error Location	Email
Description			
Created By	Harshal Alaisore 9/17/2025, 8:59 AM	Modified By	Harshal Alaisore 9/17/2025, 8:59 AM

Edit Clone

- Phone_Required (Guardian) – Ensure Guardian phone is mandatory.

SETUP > OBJECT MANAGER

Guardian

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Guardian Validation Rule

[Back to Guardian](#)

Validation Rule Detail			
Rule Name	Phone_Required	Active	☒
Error Condition Formula	ISBLANK(Phone__c)		
Error Message	Guardian phone number is required	Error Location	Phone
Description			
Created By	Harshal Alaisore 9/17/2025, 9:10 AM	Modified By	Harshal Alaisore 9/17/2025, 9:10 AM

Edit Clone

- Validate_Email_Format (Guardian) – Ensure Guardian email is valid format.

SETUP > OBJECT MANAGER

Guardian

Details

Fields & Relationships

Page Layouts

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Buttons, Links, and Actions

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Guardian Validation Rule

[Back to Guardian](#)

Help for this Page

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Validate_Email_Format	Active	✓
Error Condition Formula	NOT(ISBLANK(Email__c)) && NOT(CONTAINS(Email__c,"@"))		
Error Message	Enter a valid email address	Error Location	Email
Description			
Created By	Harshal Alaspure 9/17/2025, 9:26 AM	Modified By	Harshal Alaspure 9/17/2025, 9:26 AM

[Edit](#) [Clone](#)

- Course_Required (Enrollment) – Enrollment must always have a Course.

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

Page Layouts

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Buttons, Links, and Actions

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Enrollment Validation Rule

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Help for this Page

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Course_Required	Active	✓
Error Condition Formula	ISBLANK(Course__c)		
Error Message	Enrollment must be linked to a Course	Error Location	Course
Description			
Created By	Harshal Alaspure 9/17/2025, 9:32 AM	Modified By	Harshal Alaspure 9/17/2025, 9:32 AM

[Edit](#) [Clone](#)

- Enrollment_Date_Not_Future (Enrollment) – Enrollment date cannot be in the future.

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

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Enrollment Validation Rule

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Help for this Page

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Enrollment_Date_Not_Future	Active	✓
Error Condition Formula	AND(NOT(ISBLANK(Enrollment_Date__c)), Enrollment_Date__c > TODAY())		
Error Message	Date cannot be in the future	Error Location	Enrollment Date
Description			
Created By	Harshal Alaspure 9/17/2025, 9:33 AM	Modified By	Harshal Alaspure 9/17/2025, 9:33 AM

[Edit](#) [Clone](#)

- No_Duplicate_Enrollment (Enrollment) – Prevent duplicate enrollments for the same course/student.

SETUP > OBJECT MANAGER
Enrollment

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers

Enrollment Validation Rule

[Back to Enrollment](#) [Help for this Page](#)

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	No_Duplicate_Enrollment	Active	☒
Error Condition Formula	AND(NOT(ISBLANK(Student__C)), NOT(ISBLANK(Course__C)), ISPICKVAL(Status__C, "Active"), NOT(ISNEW()), Student__C = PRIORVALUE(Student__C), Course__C = PRIORVALUE(Course__C))		
Error Message	This student is already enrolled in the selected course	Error Location	Student
Description			
Created By	Harshal Alaspure 9/17/2025, 9:34 AM	Modified By	Harshal Alaspure 9/17/2025, 9:34 AM

[Edit](#) [Clone](#)

- Status_Required (Enrollment) – Status is mandatory on Enrollment.

SETUP > OBJECT MANAGER
Enrollment

Details
Fields & Relationships
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Enrollment Validation Rule

[Back to Enrollment](#) [Help for this Page](#)

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Status_Required	Active	☒
Error Condition Formula	ISBLANK(TEXT(Status__C))		
Error Message	Please select an Enrollment Status (Active / Completed / Dropped)	Error Location	Status
Description			
Created By	Harshal Alaspure 9/17/2025, 9:36 AM	Modified By	Harshal Alaspure 9/17/2025, 9:36 AM

[Edit](#) [Clone](#)

- Attendance_Date_Not_Future (Attendance) – Attendance date cannot be in the future.

SETUP > OBJECT MANAGER

Attendance

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

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Restriction Rules

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Triggers

Attendance Validation Rule

[Back to Attendance](#)

[Help for this Page](#)

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Attendance_Date_Not_Future	Active	✓
Error Condition Formula	AND(NOT(ISBLANK(Date__c)), Date__c > TODAY())		
Error Message	Attendance Date cannot be in the future	Error Location	Date
Description			
Created By	Harshal Aislaque: 9/17/2025, 9:40 AM	Modified By	Harshal Aislaque: 9/17/2025, 9:40 AM

[Edit](#) [Clone](#)

- Attendance_Not_Before_EnrollmentStart (Attendance) – Attendance cannot be before the enrollment date.

SETUP > OBJECT MANAGER

Attendance

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

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Restriction Rules

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Attendance Validation Rule

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[Help for this Page](#)

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Attendance_Not_Before_EnrollmentStart	Active	✓
Error Condition Formula	AND(NOT(ISBLANK(Date__c)), NOT(ISBLANK(Enrollment__r Enrollment_Date__c)), Date__c < Enrollment__r Enrollment_Date__c)		
Error Message	Attendance cannot be before the student's enrollment start date	Error Location	Date
Description			
Created By	Harshal Aislaque: 9/17/2025, 9:50 AM	Modified By	Harshal Aislaque: 9/17/2025, 9:50 AM

[Edit](#) [Clone](#)

- Enrollment_Link_Required (Attendance) – Attendance must be linked to an Enrollment record.

SETUP > OBJECT MANAGER

Attendance

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Attendance Validation Rule

[Back to Attendance](#)

[Help for this Page](#)

Validation Rule Detail [Edit](#) [Clone](#)

Rule Name	Enrollment_Link_Required	Active	✓
Error Condition Formula	ISBLANK(Enrollment__c)		
Error Message	Attendance must be linked to a valid Enrollment record	Error Location	Enrollment
Description			
Created By	Harshal Aislaque: 9/17/2025, 9:42 AM	Modified By	Harshal Aislaque: 9/17/2025, 9:42 AM

[Edit](#) [Clone](#)

- Status_Required (Attendance) – Status is mandatory on Attendance.

SETUP > OBJECT MANAGER
Attendance

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Triggers

Attendance Validation Rule

[Back to Attendance](#) [Help for this Page](#)

Validation Rule Detail		Edit		Clone	
Rule Name	Status_Required	Active	☒		
Error Condition Formula	ISBLANK(TEXT(Status__c))				
Error Message	Attendance status must be selected (Present, Absent, or Leave)		Error Location	Status	
Description					
Created By	Harshal Alaspure 9/17/2025, 9:47 AM		Modified By	Harshal Alaspure 9/17/2025, 9:47 AM	

[Edit](#) [Clone](#)

- Marks_Not_Exceed_Max (Grade) – Marks should not exceed maximum marks.

SETUP > OBJECT MANAGER
Grade

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers

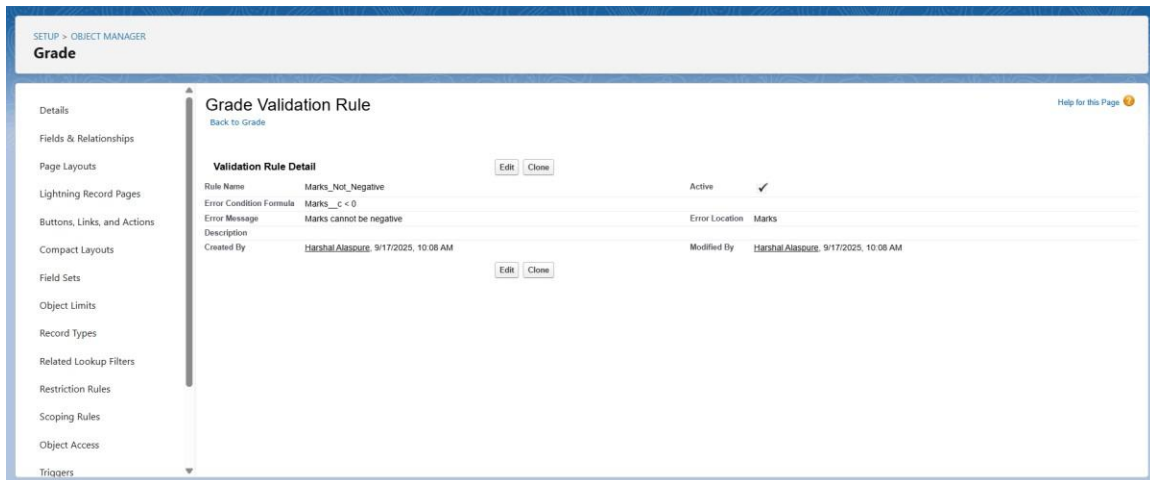
Grade Validation Rule

[Back to Grade](#) [Help for this Page](#)

Validation Rule Detail		Edit		Clone	
Rule Name	Marks_Not_Exceed_Max	Active	☒		
Error Condition Formula	AND(NOT(ISBLANK(Marks__c)), NOT(ISBLANK(Max_Marks__c)), Marks__c > Max_Marks__c)				
Error Message	Marks Obtained cannot exceed Assignment Maximum Marks		Error Location	Marks	
Description					
Created By	Harshal Alaspure 9/17/2025, 10:13 AM		Modified By	Harshal Alaspure 9/17/2025, 10:13 AM	

[Edit](#) [Clone](#)

- Marks_Not_Negative (Grade) – Marks cannot be negative.



2. Process Builder

Process Builder was used to send a welcome email when a Student record is created.

Process Name: Welcome_Student_Email

Versions: All versions are documented with screenshots.

Steps:

1. Setup → Process Builder.
2. Click New → Enter Process Name.
3. Choose 'A record changes' as the trigger.
4. Select Object: Student.
5. Define Criteria: When record is created.
6. Add Action → Email Alert → Send Welcome Email.
7. Save & Activate.

Go with the flow! With Flow Builder, the future of low-code automation, you can do everything you do with Process Builder—and more! Salesforce plans to retire Process Builder and recommends building automation in Flow Builder. [Try Flow Builder](#) | [Use Migrate to Flow Tool](#)

Process Builder - Welcome_Student_Email

Expand All Collapse All View All Processes Clone View Properties Deactivate Read Only

START

Student

Has Email

TRUE

IMMEDIATE ACTIONS

Send Welcome Email

+ Add Action

TIME

SCHEDULED ACTIONS

Set Schedule

+ Add Action

FALSE

+ Add Criteria

TRUE

IMMEDIATE ACTIONS

+ Add Action

TIME

SCHEDULED ACTIONS

Set Schedule

Define Criteria for this Action Group

Criteria Name *

Has Email

Criteria for Executing Actions *

Conditions are met

Formula evaluates to true

No criteria—just execute the actions!

Set Conditions

Field *	Operator *	Type *	Value *
1 [Student_c].Em...	Is null	Boolean	False

Conditions *

All of the conditions are met (AND)

Cancel

Process Builder - Welcome_Student_Email

Expand All Collapse All View All Processes Clone View Properties Deactivate Read Only

START

Student

Has Email

TRUE

IMMEDIATE ACTIONS

Send Welcome Email

+ Add Action

TIME

SCHEDULED ACTIONS

STOP

FALSE

+ Add Criteria

TRUE

IMMEDIATE ACTIONS

+ Add Action

TIME

SCHEDULED ACTIONS

STOP

Email Alerts

Action Name *

Send Welcome Email

Email Alert *

Send_Welcome_Email_to_new_Stu...

Cancel

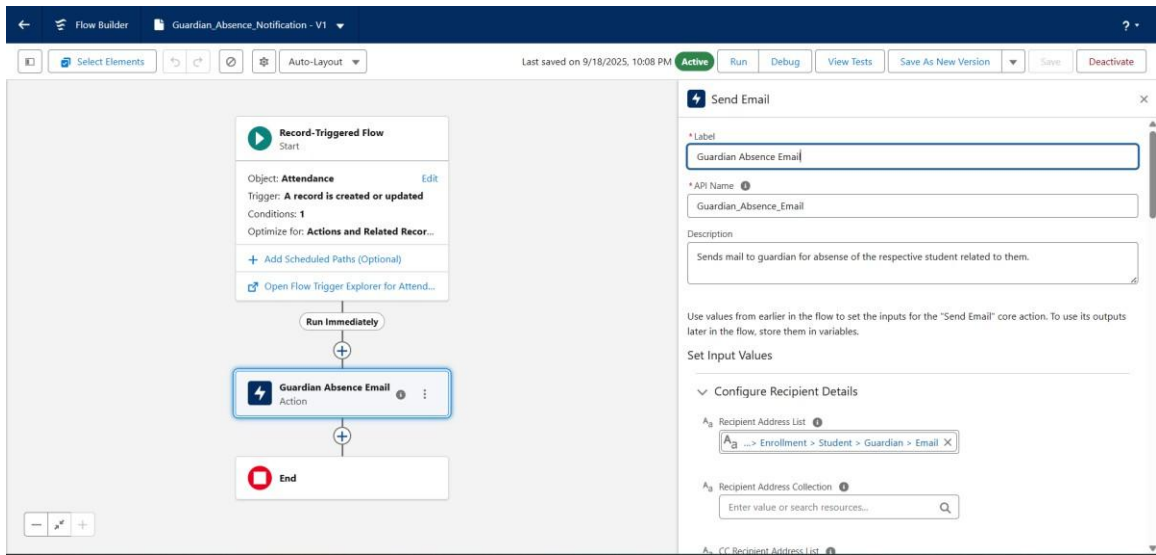
3. Flow Builder

Flows were built to automate notifications for attendance and guardian communication.

Flows Created:

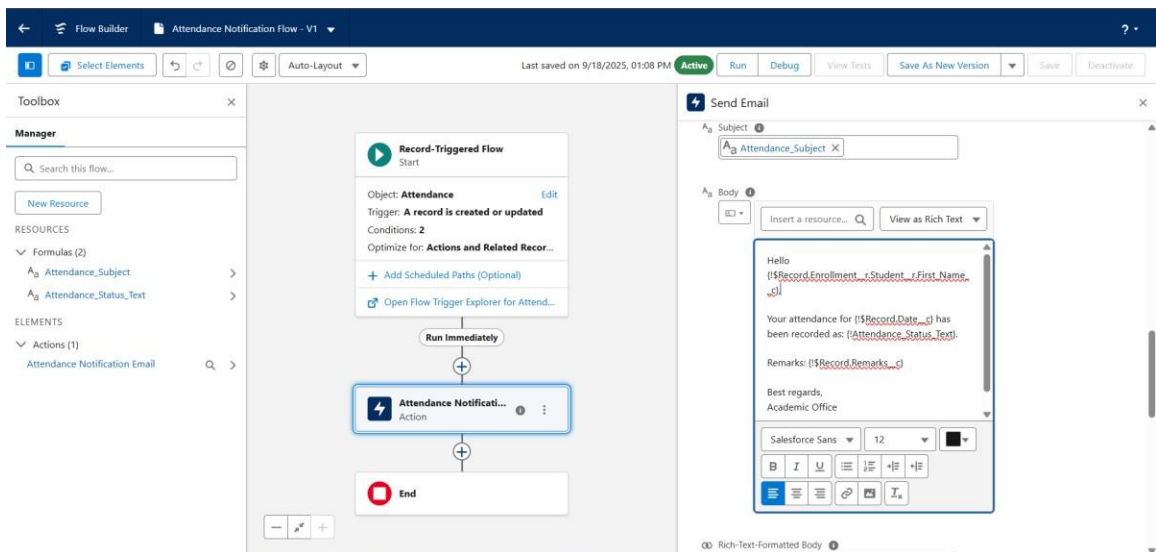
1. Guardian_Absence_Notification (Autolaunched Flow)

- Trigger: Attendance record created with Status = Absent.
- Action: Send Email to Guardian.



2. Attendance Notification Flow (Autolaunched Flow)

- Trigger: Attendance record created or updated.
- Action: Notify Student and Guardian via Email Alert.



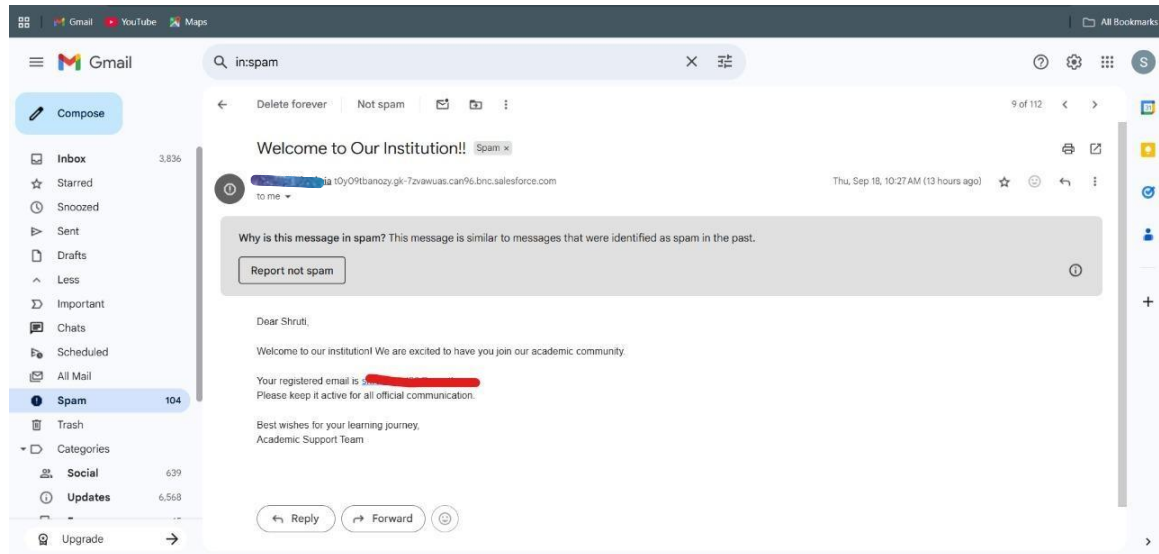
Steps to Create a Flow:

1. Setup → Flow Builder.
2. Click New Flow → Select Record-Triggered Flow.
3. Choose Object: Attendance.
4. Define Entry Conditions (e.g., Status = Absent).
5. Add Action → Email Alert.
6. Save & Activate.

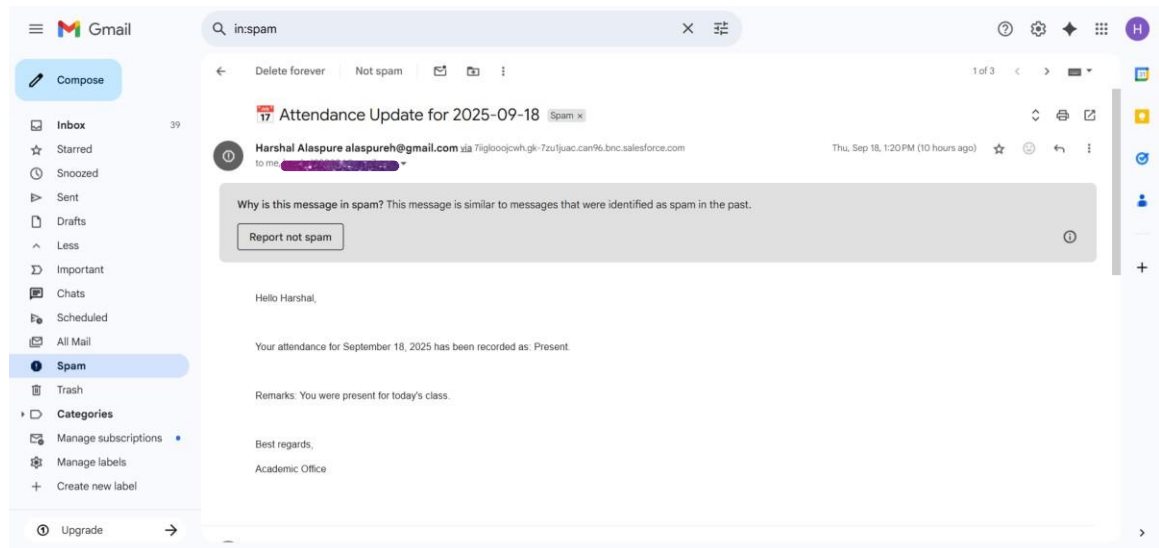
4. Email Alerts

Email Alerts were configured and linked to Process Builder and Flows. They will be shown in screenshots separately. Created alerts include:

- Student Welcome Email



- Attendance Update Notification Email



Phase 5 Apex Programming (Developer)

1. Classes & Objects

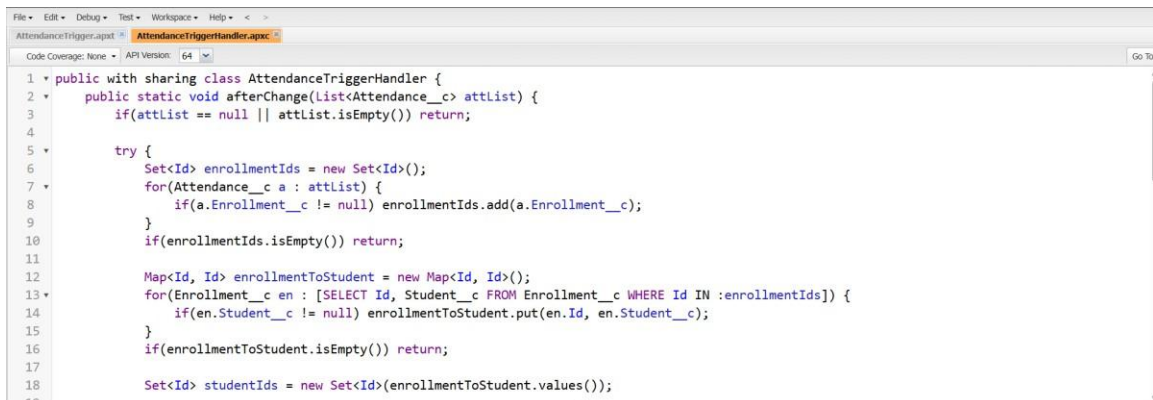
Apex classes were created to handle business logic and calculations for the student management system. Each class encapsulates specific functionality.

Steps to Create Apex Class:

1. Developer Console → File → New → Apex Class.
2. Enter Class Name and click OK.
3. Write Apex code implementing the required logic.
4. Save the class.

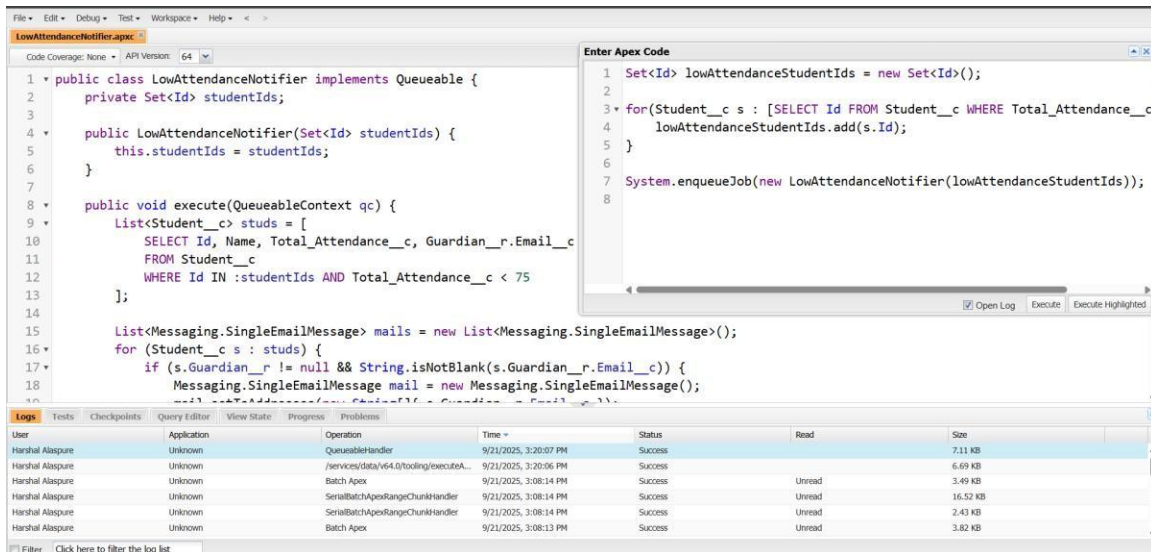
Classes Created:

- AttendanceTriggerHandler: Calculates student attendance percentage and updates the Student__c object.



```
1 public with sharing class AttendanceTriggerHandler {
2     public static void afterChange(List<Attendance__c> attList) {
3         if(attList == null || attList.isEmpty()) return;
4
5         try {
6             Set<Id> enrollmentIds = new Set<Id>();
7             for(Attendance__c a : attList) {
8                 if(a.Enrollment__c != null) enrollmentIds.add(a.Enrollment__c);
9             }
10            if(enrollmentIds.isEmpty()) return;
11
12            Map<Id, Id> enrollmentToStudent = new Map<Id, Id>();
13            for(Enrollment__c en : [SELECT Id, Student__c FROM Enrollment__c WHERE Id IN :enrollmentIds]) {
14                if(en.Student__c != null) enrollmentToStudent.put(en.Id, en.Student__c);
15            }
16            if(enrollmentToStudent.isEmpty()) return;
17
18            Set<Id> studentIds = new Set<Id>(enrollmentToStudent.values());
```

- LowAttendanceNotifier: Sends email alerts to guardians if attendance falls below threshold.



The screenshot displays the Salesforce IDE with the `LowAttendanceNotifier.apex` class open. The class implements the `Queueable` interface and contains an `execute` method that queries for students with attendance below 75 and sends email alerts to their guardians. An 'Enter Apex Code' window shows a snippet of the `execute` method. Below the code editor, the 'Logs' tab is active, showing a table of execution logs for the class.

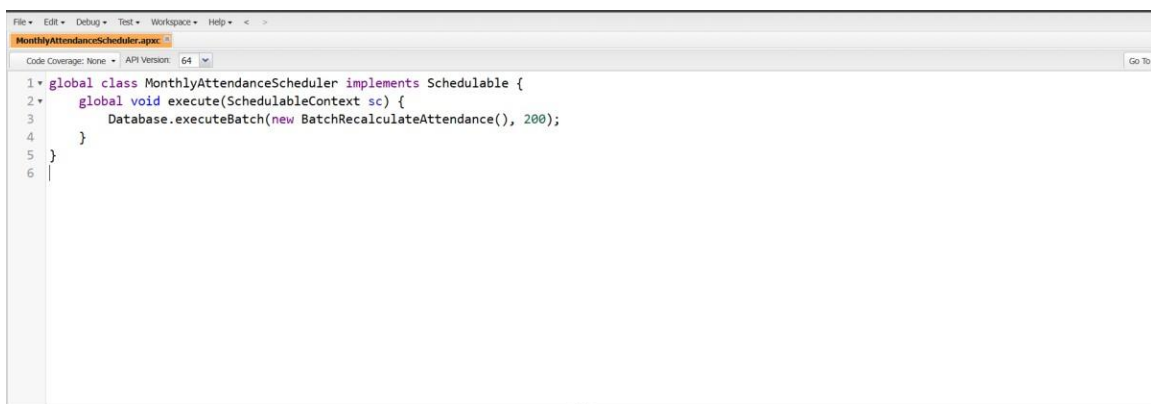
User	Application	Operation	Time	Status	Read	Size
Harshal Alaspure	Unknown	QueueableHandler	9/21/2025, 3:20:07 PM	Success		7.11 KB
Harshal Alaspure	Unknown	/services/data/v64.0/tooling/execute...	9/21/2025, 3:20:06 PM	Success		6.69 KB
Harshal Alaspure	Unknown	Batch Apex	9/21/2025, 3:08:14 PM	Success	Unread	3.49 KB
Harshal Alaspure	Unknown	SerialBatchApexRangeChunkHandler	9/21/2025, 3:08:14 PM	Success	Unread	16.52 KB
Harshal Alaspure	Unknown	SerialBatchApexRangeChunkHandler	9/21/2025, 3:08:14 PM	Success	Unread	2.43 KB
Harshal Alaspure	Unknown	Batch Apex	9/21/2025, 3:08:13 PM	Success	Unread	3.82 KB

- BatchRecalculateAttendance: Recalculates attendance percentages for all students in bulk.



The screenshot shows the `BatchRecalculateAttendance.apex` class in the Salesforce IDE. This class implements the `Database.Batchable<Object>` interface. It includes a `start` method to get the query locator, an `execute` method to process a scope of students, and a `finish` method to log the results. The `execute` method queries for attendance records for the given scope of students and calculates the attendance percentage.

- MonthlyAttendanceScheduler: Schedules monthly batch execution to recalculate attendance.



The screenshot displays the `MonthlyAttendanceScheduler.apex` class in the Salesforce IDE. This class implements the `Schedulable` interface and contains an `execute` method that triggers the `BatchRecalculateAttendance` batch class to recalculate attendance for all students.

- - ErrorLogger – Logs exceptions to Error_Log__c for debugging and auditing.

```

1 public with sharing class ErrorLogger {
2     public static void log(Exception ex, String context, Id relatedId) {
3         try {
4             if (ex == null) return;
5
6             Error_Log__c logRec = new Error_Log__c();
7
8             if (context != null && context.length() > 255) {
9                 logRec.Context__c = context.substring(0, 255);
10            } else {
11                logRec.Context__c = context;
12            }
13
14            String msg = ex.getMessage();
15            if (msg != null && msg.length() > 32000) msg = msg.substring(0, 32000);
16            logRec.Exception_Message__c = msg;
17
18            String stack = ex.getStackTraceString();
19            if (stack != null && stack.length() > 32000) stack = stack.substring(0, 32000);
20            logRec.Stack_Trace__c = stack;
21        } catch (Exception e) {
22            // Log the exception
23        }
24    }
25 }

```

- - Error Log Object was created same as mentioned steps in Phase 3 for this Phase

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Context	Context__c	Text(255)		
Created By	CreatedById	Lookup(User)		
ErrorLogNumber	Name	Auto Number		✓
Exception Message	Exception_Message__c	Long Text Area(32000)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Related Record Id	Related_Record_Id__c	Text(18)		
Stack Trace	Stack_Trace__c	Long Text Area(32000)		

2. Apex Triggers

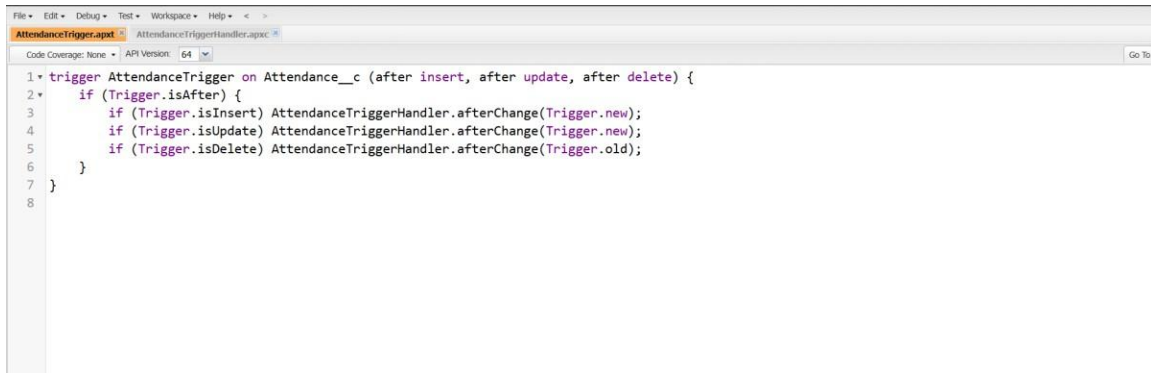
-AttendanceTrigger

Object : Attendance__c

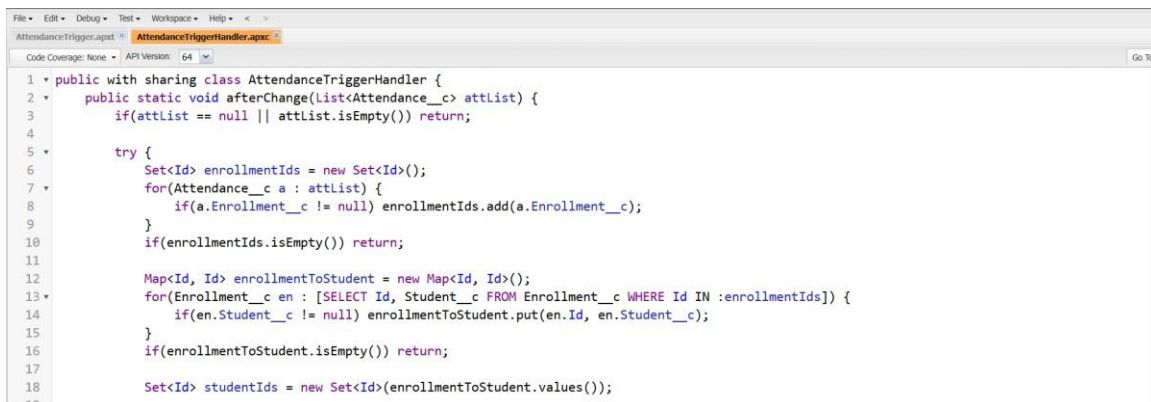
Trigger was created to automatically update attendance percentages when Attendance__c records are inserted, updated, or deleted.

Trigger: AttendanceTrigger on Attendance__c (after insert, update, delete)

- Calls AttendanceTriggerHandler.afterChange() to update attendance percentages.



```
1 trigger AttendanceTrigger on Attendance__c (after insert, after update, after delete) {
2     if (Trigger.isAfter) {
3         if (Trigger.isInsert) AttendanceTriggerHandler.afterChange(Trigger.new);
4         if (Trigger.isUpdate) AttendanceTriggerHandler.afterChange(Trigger.new);
5         if (Trigger.isDelete) AttendanceTriggerHandler.afterChange(Trigger.old);
6     }
7 }
8
```



```
1 public with sharing class AttendanceTriggerHandler {
2     public static void afterChange(List<Attendance__c> attlist) {
3         if(attlist == null || attlist.isEmpty()) return;
4
5         try {
6             Set<Id> enrollmentIds = new Set<Id>();
7             for(Attendance__c a : attlist) {
8                 if(a.Enrollment__c != null) enrollmentIds.add(a.Enrollment__c);
9             }
10            if(enrollmentIds.isEmpty()) return;
11
12            Map<Id, Id> enrollmentToStudent = new Map<Id, Id>();
13            for(Enrollment__c en : [SELECT Id, Student__c FROM Enrollment__c WHERE Id IN :enrollmentIds]) {
14                if(en.Student__c != null) enrollmentToStudent.put(en.Id, en.Student__c);
15            }
16            if(enrollmentToStudent.isEmpty()) return;
17
18            Set<Id> studentIds = new Set<Id>(enrollmentToStudent.values());
19        }
20    }
21 }
```

Steps to Create Trigger:

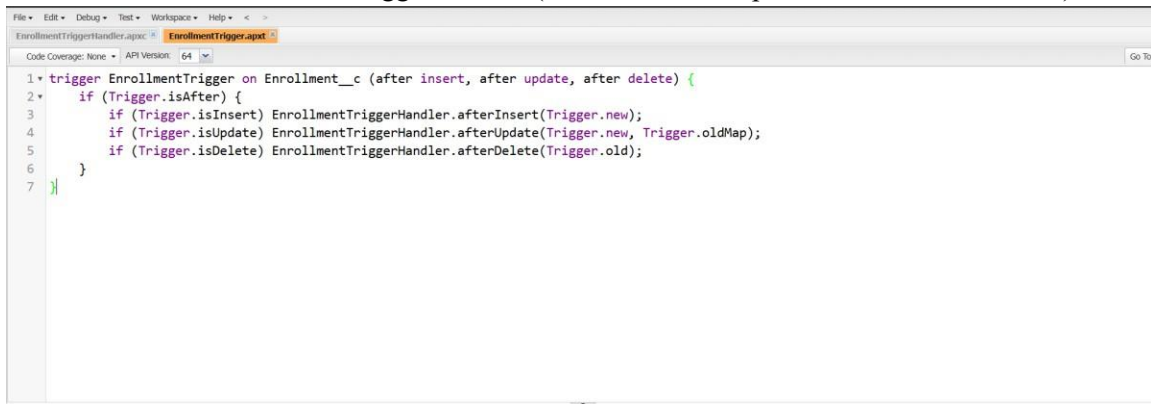
1. Developer Console → File → New → Apex Trigger.
2. Select Object: Attendance__c.
3. Write Trigger code calling the handler.
4. Save the trigger.

-EnrollmentTrigger

Object: Enrollment__c

Purpose: Handle automation related to enrollments, e.g., updating related records or sending notifications when a student is enrolled, updated, or removed from a course.

Handler Called: EnrollmentTriggerHandler (afterInsert, afterUpdate, afterDelete methods)



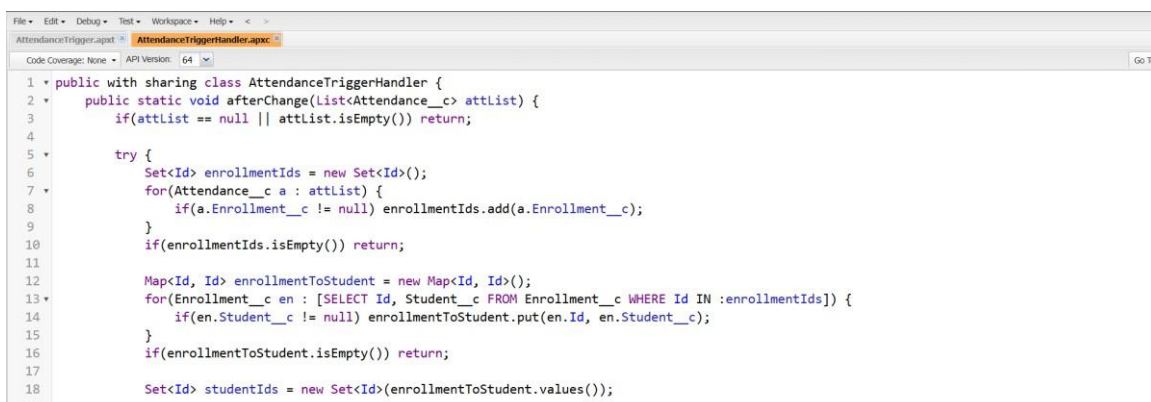
Steps to Create Trigger:

1. Developer Console → File → New → Apex Trigger.
2. Select Object: Enrollment__c.
3. Write trigger code calling the handler:

3. Trigger Design Pattern

- Trigger handler classes (e.g. AttendanceTriggerHandler) were implemented to keep trigger logic separate from the trigger definition.
- This ensures reusability and easier maintenance.

AttendanceTriggerHandler Screenshot



EnrollmentTriggerHandler Screenshot

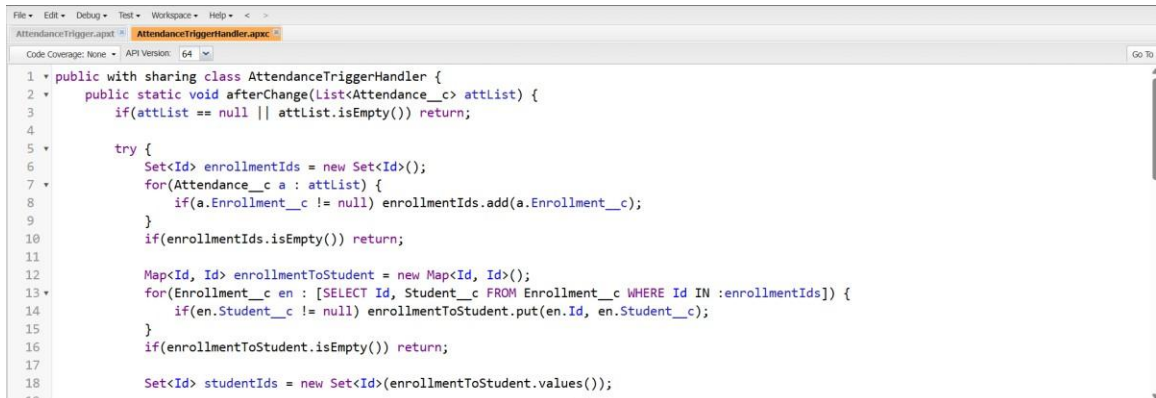


```
1 public with sharing class EnrollmentTriggerHandler {
2
3     public static void afterInsert(List<Enrollment__c> newList) {
4         updateStudentCourseCounts(newList);
5     }
6
7     public static void afterUpdate(List<Enrollment__c> newList, Map<Id, Enrollment__c> oldMap) {
8         updateStudentCourseCounts(newList);
9     }
10
11     public static void afterDelete(List<Enrollment__c> oldList) {
12         updateStudentCourseCounts(oldList);
13     }
14
15     private static void updateStudentCourseCounts(List<Enrollment__c> enrolls) {
16         try {
17             Set<Id> studentIds = new Set<Id>();
18             for (Enrollment__c e : enrolls) {
19                 if (e.Student__c != null) studentIds.add(e.Student__c);
20             }
21             // Update student course counts
22         } catch (Exception ex) {
23             // Handle exception
24         }
25     }
26 }
```

The screenshot shows an IDE window with the file 'EnrollmentTriggerHandler.apex' open. The code is written in Apex and defines a class 'EnrollmentTriggerHandler' with four static methods: 'afterInsert', 'afterUpdate', 'afterDelete', and 'updateStudentCourseCounts'. The 'updateStudentCourseCounts' method is private and contains logic to update student course counts based on a list of enrollment records. The IDE interface includes a menu bar (File, Edit, Debug, Test, Workspace, Help), a toolbar with 'Code Coverage: None' and 'API Version: 64', and a bottom panel with tabs for 'Logs', 'Tests', 'Checkpoints', 'Query Editor', 'View State', 'Progress', and 'Problems'.

4. SOQL & SOSL

- SOQL used in AttendanceTriggerHandler and BatchRecalculateAttendance to fetch Enrollment and Attendance records.
- AggregateResult used to calculate totals and present counts.



```
1 public with sharing class AttendanceTriggerHandler {
2     public static void afterChange(List<Attendance__c> attlist) {
3         if(attlist == null || attlist.isEmpty()) return;
4
5         try {
6             Set<Id> enrollmentIds = new Set<Id>();
7             for(Attendance__c a : attlist) {
8                 if(a.Enrollment__c != null) enrollmentIds.add(a.Enrollment__c);
9             }
10            if(enrollmentIds.isEmpty()) return;
11
12            Map<Id, Id> enrollmentToStudent = new Map<Id, Id>();
13            for(Enrollment__c en : [SELECT Id, Student__c FROM Enrollment__c WHERE Id IN :enrollmentIds]) {
14                if(en.Student__c != null) enrollmentToStudent.put(en.Id, en.Student__c);
15            }
16            if(enrollmentToStudent.isEmpty()) return;
17
18            Set<Id> studentIds = new Set<Id>(enrollmentToStudent.values());
```



```
1 global class BatchRecalculateAttendance implements Database.Batchable<sObject> {
2     global Database.QueryLocator start(Database.BatchableContext bc) {
3         System.debug('Batch started...');
4         return Database.getQueryLocator([SELECT Id FROM Student__c]);
5     }
6
7     global void execute(Database.BatchableContext bc, List<Student__c> scope) {
8         System.debug('Processing students: ' + scope);
9
10        Set<Id> studentIds = new Set<Id>();
11        for (Student__c s : scope) studentIds.add(s.Id);
12
13        List<Attendance__c> attlist = [
14            SELECT Id, Status__c, Enrollment__r.Student__c
15            FROM Attendance__c
16            WHERE Enrollment__r.Student__c IN :studentIds
17        ];
18        System.debug('Attendance records found: ' + attlist.size());
```

5. Collections: List, Set, Map

- List used to batch update students.
- Set used to store unique Enrollment and Student IDs.
- Map<Id, Id> used to map Enrollment to Student.
- Map<Id, Decimal> used to store attendance percentages.

6. Control Statements

- if, for loops, try-catch used throughout handler and batch classes for flow control and error handling.

7. Batch Apex

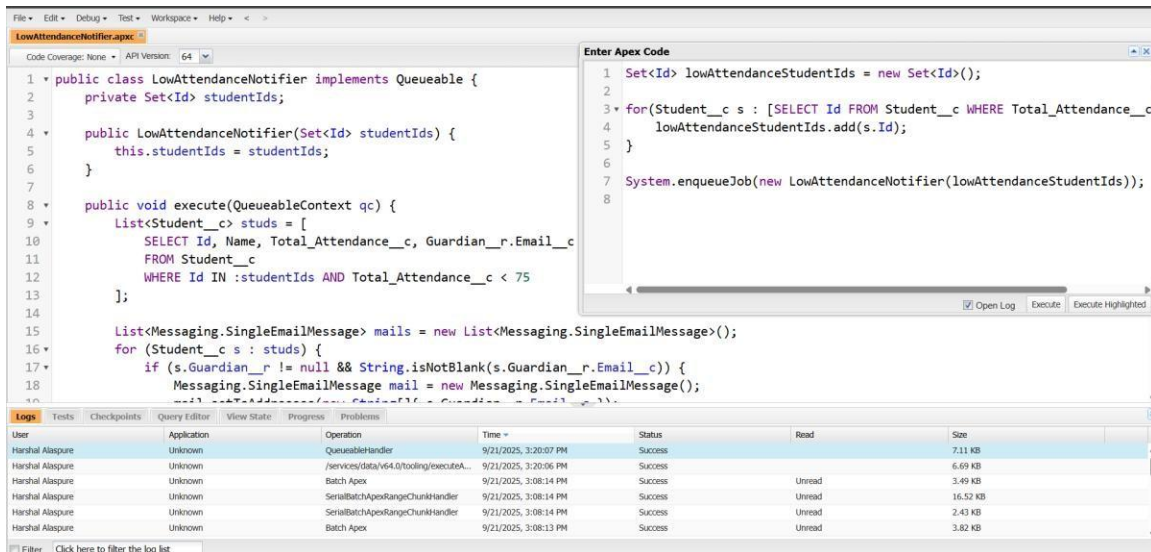
- BatchRecalculateAttendance class implements Database.Batchable to recalculate attendance for all students.
- Developer Console → Execute Anonymous to run batch manually.



```
1 global class BatchRecalculateAttendance implements Database.Batchable<Object> {
2     global Database.QueryLocator start(Database.BatchableContext bc) {
3         System.debug('Batch started...');
4         return Database.getQueryLocator([SELECT Id FROM Student__c]);
5     }
6
7     global void execute(Database.BatchableContext bc, List<Student__c> scope) {
8         System.debug('Processing students: ' + scope);
9
10        Set<Id> studentIds = new Set<Id>();
11        for (Student__c s : scope) studentIds.add(s.Id);
12
13        List<Attendance__c> attList = [
14            SELECT Id, Status__c, Enrollment__r.Student__c
15            FROM Attendance__c
16            WHERE Enrollment__r.Student__c IN :studentIds
17        ];
18        System.debug('Attendance records found: ' + attList.size());
19    }
20 }
```

8. Queueable Apex

- LowAttendanceNotifier implements Queueable to send asynchronous email notifications to guardians for low attendance.
- Accepts a Set of Student IDs and processes in background.



```
1 public class LowAttendanceNotifier implements Queueable {
2     private Set<Id> studentIds;
3
4     public LowAttendanceNotifier(Set<Id> studentIds) {
5         this.studentIds = studentIds;
6     }
7
8     public void execute(QueueableContext qc) {
9         List<Student__c> studs = [
10             SELECT Id, Name, Total_Attendance__c, Guardian__r.Email__c
11             FROM Student__c
12             WHERE Id IN :studentIds AND Total_Attendance__c < 75
13         ];
14
15         List<Messaging.SingleEmailMessage> mails = new List<Messaging.SingleEmailMessage>();
16         for (Student__c s : studs) {
17             if (s.Guardian__r != null && String.isNotBlank(s.Guardian__r.Email__c)) {
18                 Messaging.SingleEmailMessage mail = new Messaging.SingleEmailMessage();
19                 mail.setToAddresses(new String[] { s.Guardian__r.Email__c });
20                 mail.setSubject('Low Attendance Notification');
21                 mail.setBody('Your child ' + s.Name + ' has a low attendance of ' + s.Total_Attendance__c);
22                 mails.add(mail);
23             }
24         }
25         Messaging.sendEmail(mails);
26     }
27 }
```

Enter Apex Code

```
1 Set<Id> lowAttendanceStudentIds = new Set<Id>();
2
3 for(Student__c s : [SELECT Id FROM Student__c WHERE Total_Attendance__c < 75])
4     lowAttendanceStudentIds.add(s.Id);
5
6
7 System.enqueueJob(new LowAttendanceNotifier(lowAttendanceStudentIds));
8
```

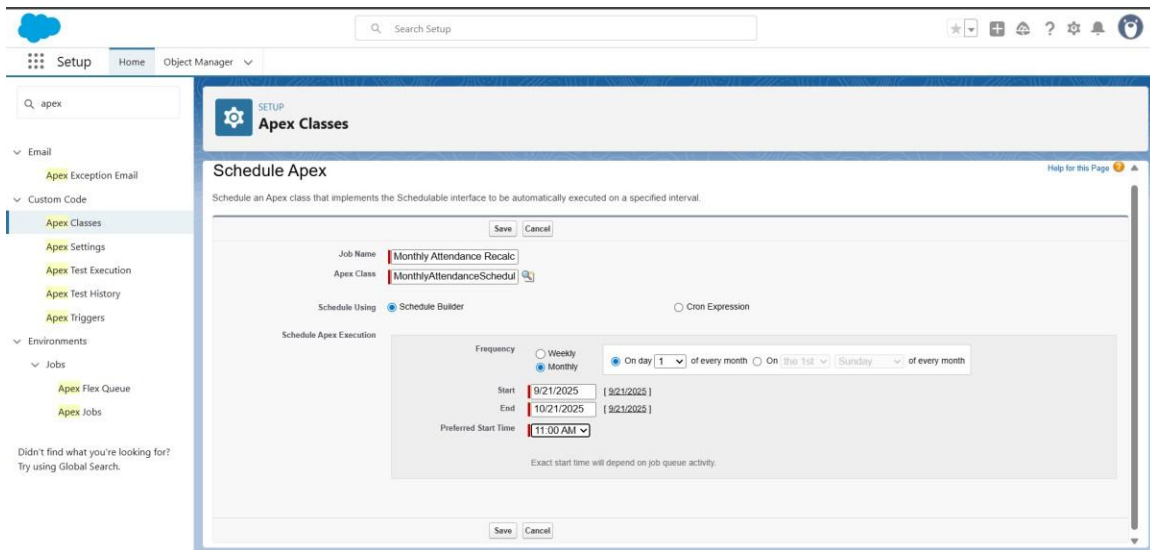
User	Application	Operation	Time	Status	Read	Size
Harshal Alaspure	Unknown	QueueableHandler	9/21/2025, 3:20:07 PM	Success		7.11 KB
Harshal Alaspure	Unknown	/services/data/v64.0/tooling/execute...	9/21/2025, 3:20:06 PM	Success		6.69 KB
Harshal Alaspure	Unknown	Batch Apex	9/21/2025, 3:08:14 PM	Success	Unread	3.49 KB
Harshal Alaspure	Unknown	SerialBatchApexRangeChunkHandler	9/21/2025, 3:08:14 PM	Success	Unread	16.52 KB
Harshal Alaspure	Unknown	SerialBatchApexRangeChunkHandler	9/21/2025, 3:08:14 PM	Success	Unread	2.43 KB
Harshal Alaspure	Unknown	Batch Apex	9/21/2025, 3:08:13 PM	Success	Unread	3.82 KB

9. Scheduled Apex

- MonthlyAttendanceScheduler schedules BatchRecalculateAttendance to run monthly.

```
File • Edit • Debug • Test • Workspace • Help • < >
MonthlyAttendanceScheduler.apex
Code Coverage: None • API Version: 64
1 global class MonthlyAttendanceScheduler implements Schedulable {
2     global void execute(SchedulableContext sc) {
3         Database.executeBatch(new BatchRecalculateAttendance(), 200);
4     }
5 }
6
```

- Setup → Apex Classes → Schedule Apex or via Developer Console.



10. Future Methods

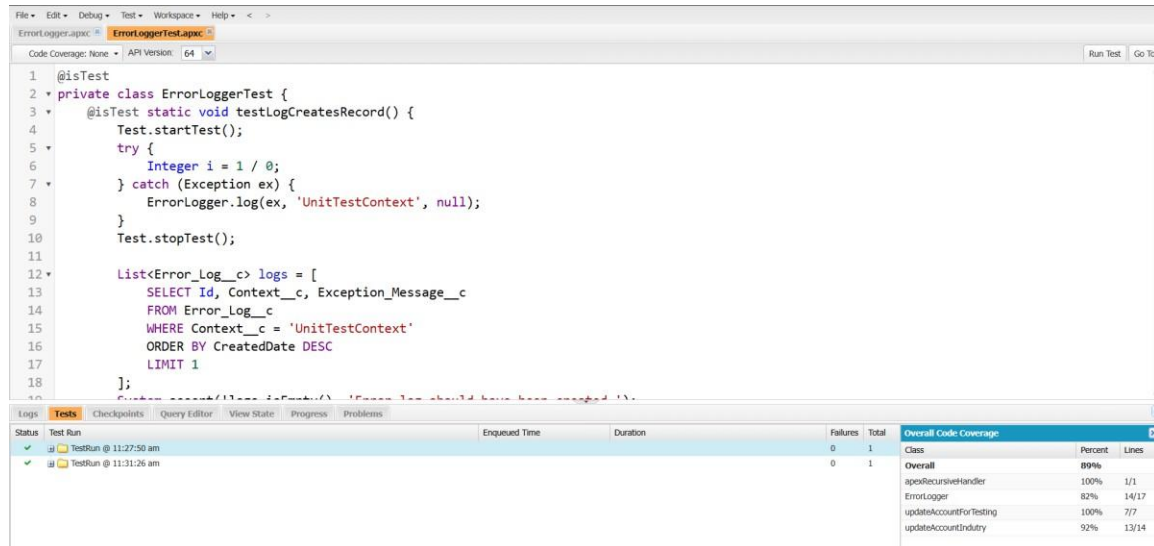
- Not implemented in this phase as Queueable Apex handled asynchronous processing.

11. Exception Handling

- try-catch blocks implemented in AttendanceTriggerHandler and Queueable classes.
- System.debug used to log exceptions for troubleshooting.

12. Test Classes

- ErrorLoggerTestClass: Tests the ErrorLogger Class and trigger working.
- Ensures code coverage > 75%.



13. Asynchronous Processing

- Batch Apex, Queueable Apex, and Scheduled Apex implemented to handle bulk data and background processing.

14. Process Flow Summazed

A. AttendanceTriggerHandler

Developer Console → File → New → Apex Class → Write Code → Save → Trigger fires on Attendance insert/update/delete → Handler calculates percentage → Update Student__c.Total_Attendance__c.

B. LowAttendanceNotifier

Developer Console → File → New → Apex Class → Write Queueable code → Save → Execute Queueable passing Set of low-attendance students → Emails sent to Guardian__c.Email__c.

C. BatchRecalculateAttendance

Developer Console → File → New → Apex Class → Write Batchable code → Save → Execute via Developer Console or Scheduled Apex → Attendance percentages recalculated for all students.

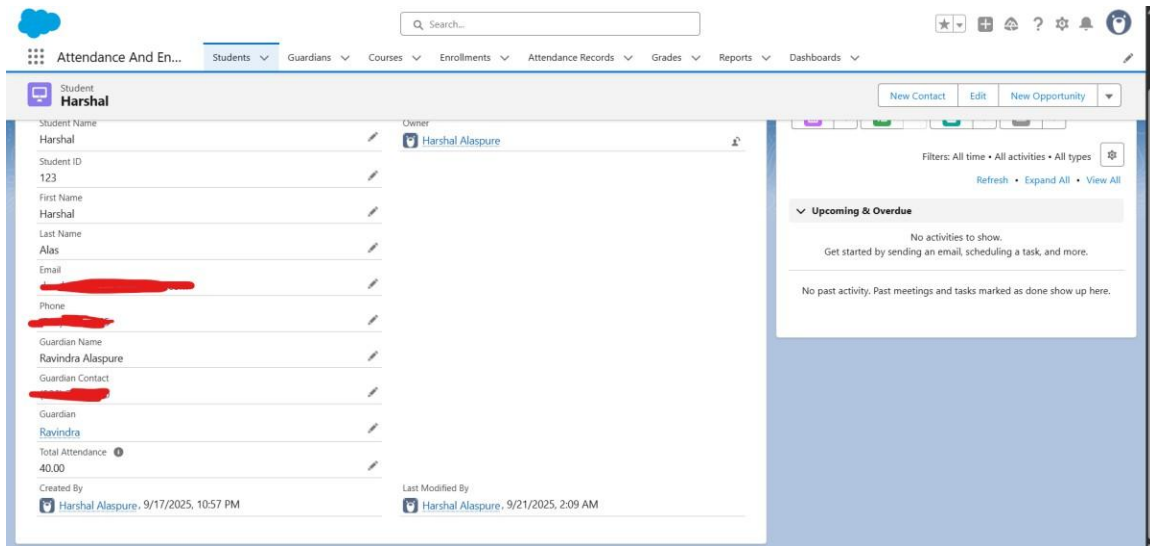
D. MonthlyAttendanceScheduler

Developer Console → File → New → Apex Class → Write Schedulable code → Save → Setup → Apex Classes → Schedule Apex → Runs monthly → Executes BatchRecalculateAttendance.

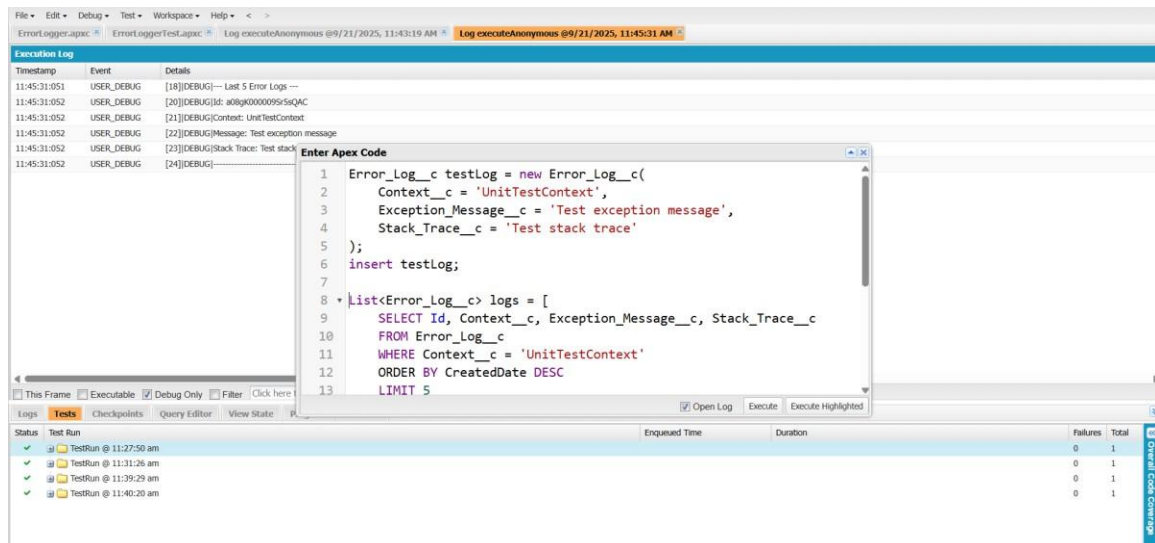
15. Test and Result Screenshots

Here are all the screenshots of the triggering outputs

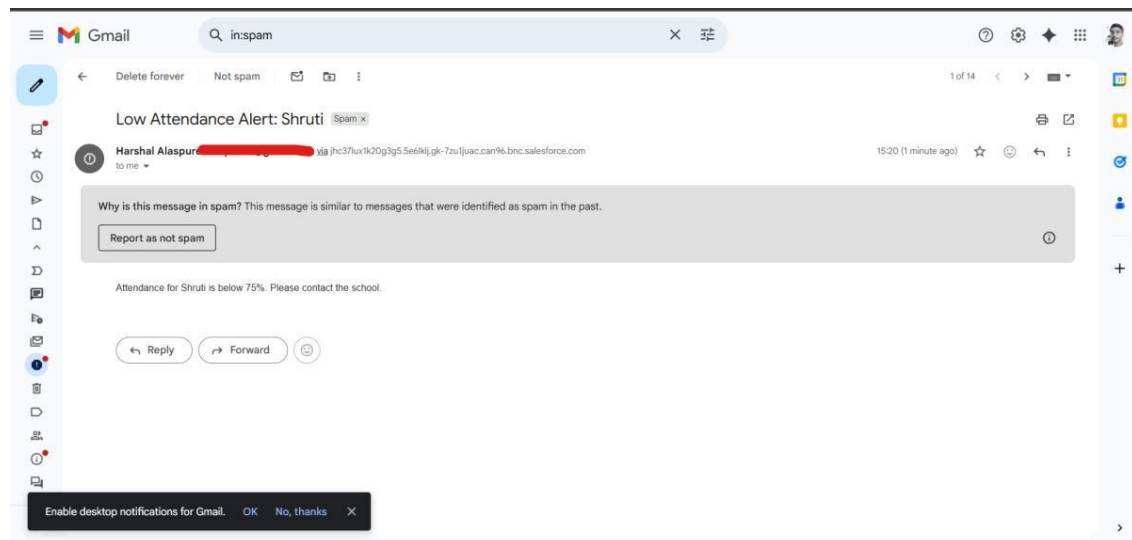
- When AttendanceTrigger, AttendanceTriggerHandler worked and I noted attendance for a few days, the “Total Attendance” field of Student_c got autofilled via triggering event.



- When ErrorLoggerClass worked and the AnonymousWindow used for taking a test on some error data, here's the whole output screenshot.



- When the queueable task was created and ran, the automated mail to the guardian email of respective student has been sent if their attendance is found below 75%.

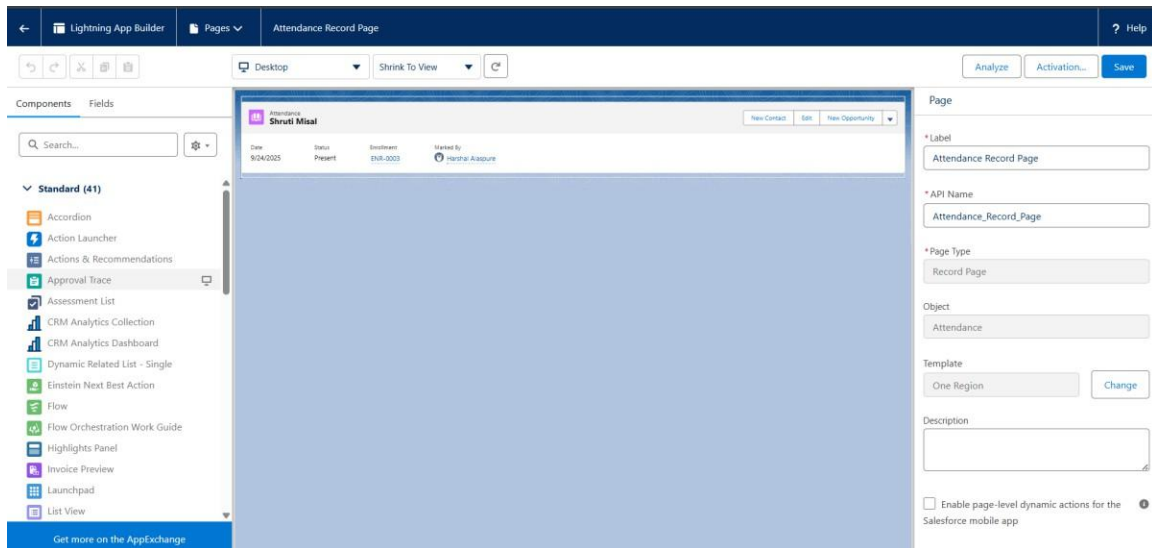


Phase 6 - 10 Till the Documentation

Phase 6: User Interface Development

Lightning App Builder -> Record Pages

Setup → Lightning App Builder → New → Record Page → Object: Attendance__c → Add Highlights Panel, Related Lists (Student, Course, Status, Date).



Tabs (Custom Object Tabs)

Setup → Tabs → New → Custom Object Tab → Attendance Records

Setup → Tabs → New → Custom Object Tab → Courses

Setup → Tabs → New → Custom Object Tab → Enrollments

Setup → Tabs → New → Custom Object Tab → Grades

Setup → Tabs → New → Custom Object Tab → Guardians

Setup → Tabs → New → Custom Object Tab → Integration Logs

Setup → Tabs → New → Custom Object Tab → Students

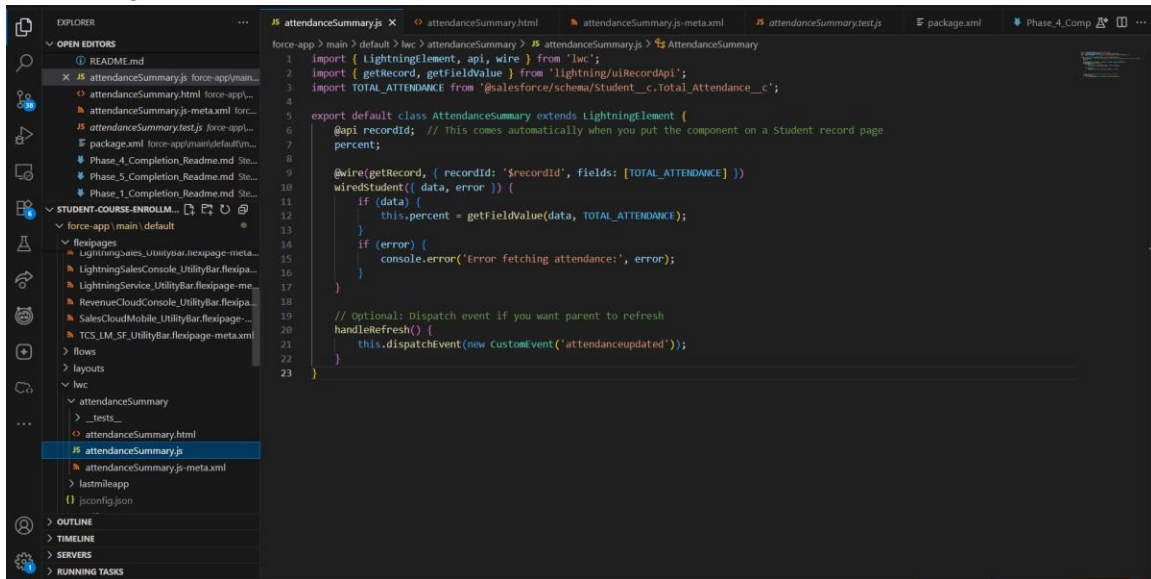
The screenshot shows the 'Edit Custom Object Tab' page for 'Attendance Records'. The left sidebar has a search bar with 'tabs' and a 'User Interface' section with 'Rename Tabs and Labels' and 'Tabs'. The main content area is titled 'Attendance Records' and contains a 'Custom Tab Definition Edit' form. The form has fields for 'Tab Label' (Attendance Records), 'Object' (Attendance), and 'Tab Style' (Books). There is also a section for 'Optional' choices, including 'Home Page Custom Link' and 'Splash Page Custom Link'. A 'Description' field is at the bottom. The page has a 'Save' button and a 'Cancel' button.

The screenshot shows the 'Custom Tabs' page in the Salesforce Setup interface. The left sidebar has a search bar with 'tabs' and a 'User Interface' section with 'Rename Tabs and Labels' and 'Tabs'. The main content area is titled 'Custom Tabs' and contains a table of 'Custom Object Tabs'. The table has columns for 'Action', 'Label', 'Tab Style', and 'Description'. The table lists several tabs: 'Attendance Records', 'Courses', 'Enrollments', 'Grades', 'Guardians', 'Integration Logs', and 'Students'. Each tab has a 'Tab Style' (e.g., Books, Bank, Airplane, Leaf, Red Cross, Stopwatch, Computer) and a 'Description' field. The page also has a 'Web Tabs' section at the bottom, which is currently empty.

Action	Label	Tab Style	Description
Edit Del	Attendance Records	Books	
Edit Del	Courses	Bank	
Edit Del	Enrollments	Airplane	
Edit Del	Grades	Leaf	
Edit Del	Guardians	Red Cross	
Edit Del	Integration Logs	Stopwatch	
Edit Del	Students	Computer	

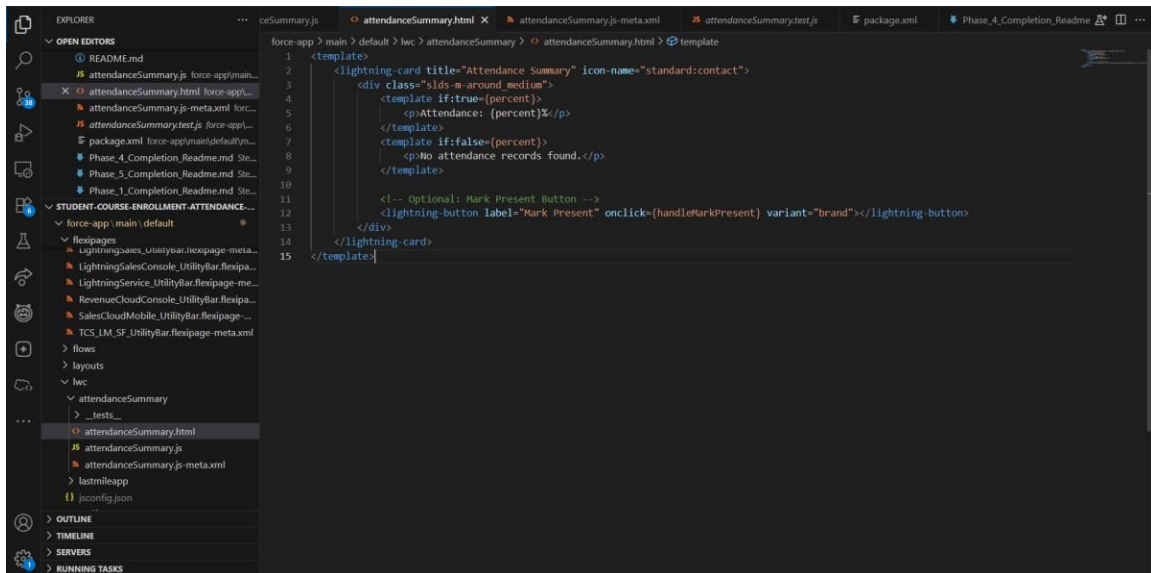
LWC (Lightning Web Components)

VS Code → Create LWC → attendanceSummary → Deploy to Salesforce → Add to Attendance Record Page.



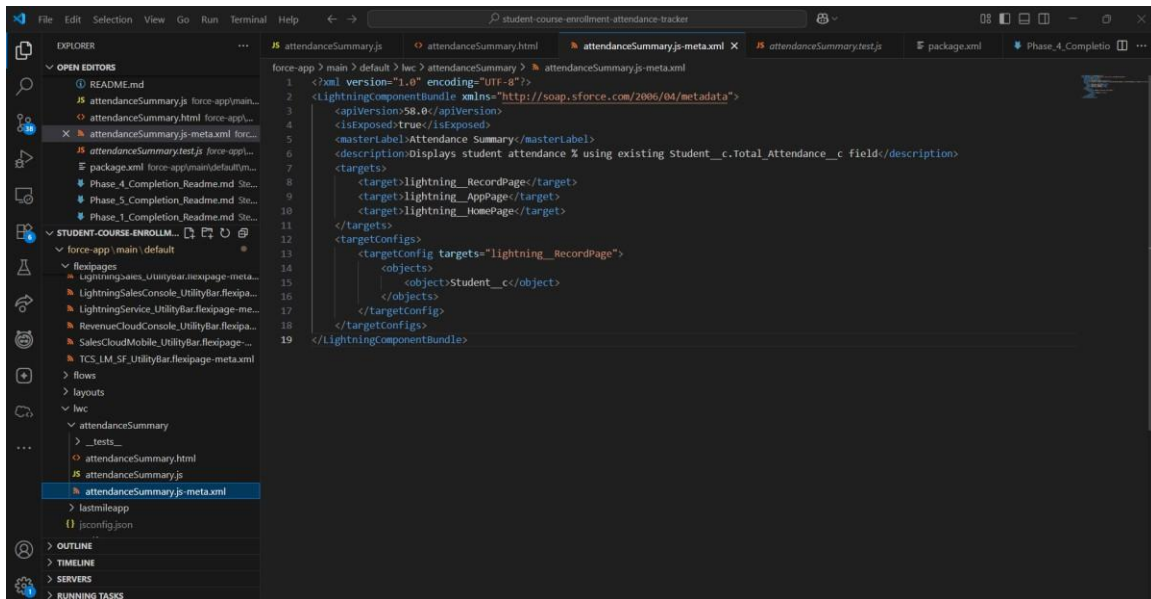
The screenshot shows the VS Code editor with the 'attendanceSummary.js' file open. The Explorer sidebar on the left shows the project structure, including the 'attendanceSummary' folder. The main editor displays the following JavaScript code:

```
force-app > main > default > lwc > attendanceSummary > JS attendanceSummary.js > AttendanceSummary
1 import { LightningElement, api, wire } from 'lwc';
2 import { getRecord, getFieldValue } from 'lightning/uiRecordApi';
3 import TOTAL_ATTENDANCE from '@salesforce/schema/Student__c.Total_Attendance__c';
4
5 export default class AttendanceSummary extends LightningElement {
6   @api recordId; // This comes automatically when you put the component on a Student record page
7   percent;
8
9   @wire(getRecord, { recordId: '$recordId', fields: [TOTAL_ATTENDANCE] })
10   wiredStudent({ data, error }) {
11     if (data) {
12       this.percent = getFieldValue(data, TOTAL_ATTENDANCE);
13     }
14     if (error) {
15       console.error('Error fetching attendance:', error);
16     }
17   }
18
19   // Optional: Dispatch event if you want parent to refresh
20   handleRefresh() {
21     this.dispatchEvent(new CustomEvent('attendanceupdated'));
22   }
23 }
```



The screenshot shows the VS Code editor with the 'attendanceSummary.html' file open. The Explorer sidebar on the left shows the project structure, including the 'attendanceSummary' folder. The main editor displays the following HTML template code:

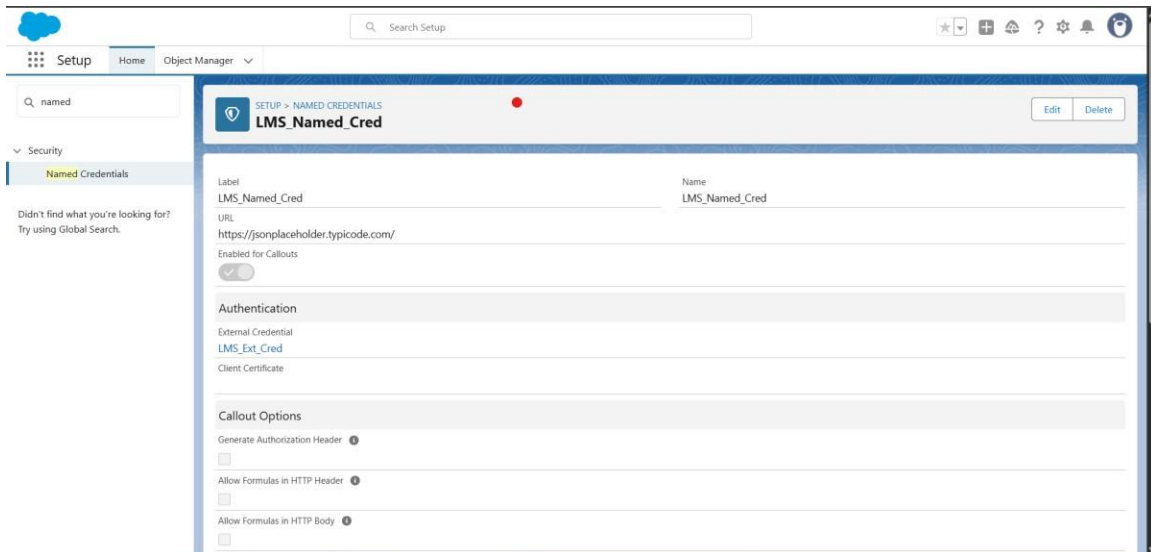
```
force-app > main > default > lwc > attendanceSummary > attendanceSummary.html > template
1 <template>
2   <lightning-card title="Attendance Summary" icon-name="standard:contact">
3     <div class="slds-m-around_medium">
4       <template if:true={percent}>
5         <p>Attendance: {percent}%</p>
6       </template>
7       <template if:false={percent}>
8         <p>No attendance records found.</p>
9       </template>
10
11       <!-- Optional: Mark Present Button -->
12       <lightning-button label="Mark Present" onclick={handleMarkPresent} variant="brand"></lightning-button>
13     </div>
14   </lightning-card>
15 </template>
```



Phase 7: Integration & External Access

Named Credentials

Setup → Named Credentials → New → Label: LMS_Named_Cred → URL:
<https://jsonplaceholder.typicode.com/> → External Credential: LMS_Ext_Cred → Save.



External Credentials

Setup → External Credentials → New → Label: LMS_Ext_Cred → Authentication Protocol: Basic Authentication → Save.

The screenshot shows the Salesforce Setup interface for managing External Credentials. The left sidebar shows the navigation menu with 'Setup' selected. The main content area is titled 'LMS_Ext_Cred' and displays the following information:

- Label:** LMS_Ext_Cred
- Authentication Protocol:** Basic Authentication
- Managed Package Access:** (Empty section)
- Created By:** Namespace
- Related Named Credentials:** A table showing one related credential: LMS_Named_Cred with URL https://jsonplaceholder.typicode.com/.
- Principals:** A table showing one principal: ReqRes_Principal, Named Principal, Configured, testuser.

The screenshot shows the Salesforce IDE with an Apex class named LmsSimpleGet. The code is as follows:

```
1 public with sharing class LmsSimpleGet {
2
3     @AuraEnabled
4     public static String getCourses() {
5         HttpRequest req = new HttpRequest();
6         req.setEndpoint('callout:LMS_Named_Cred/courses'); // relative path
7         req.setMethod('GET');
8
9         Http http = new Http();
10        HttpResponse res = http.send(req);
11
12        return res.getBody();
13    }
14 }
15 }
```

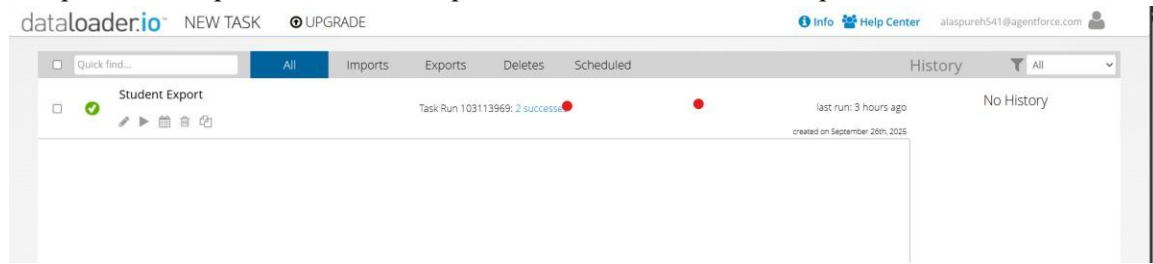
Phase 8: Data Management & Deployment

Data Loader

Open Data Loader → Login with Salesforce → Select Insert → Object: Student_c → Choose CSV file → Map fields → Finish.

Data Export & Backup

Setup → Data Export → Schedule Export → Select Student_c → Export to CSV → Download.



FileHomeInsertDrawPage LayoutFormulasDataReviewViewHelpAcrobatTell me what you want to do

PasteCutCopyFormat PainterClipboard

Calibri11AaBbUuFont

Wrap TextMerge & CenterAlignment

General\$%&Number

Conditional FormattingFormat as TableFormat as CellInsertDeleteFormatCells

AutoSumFillClearEditing

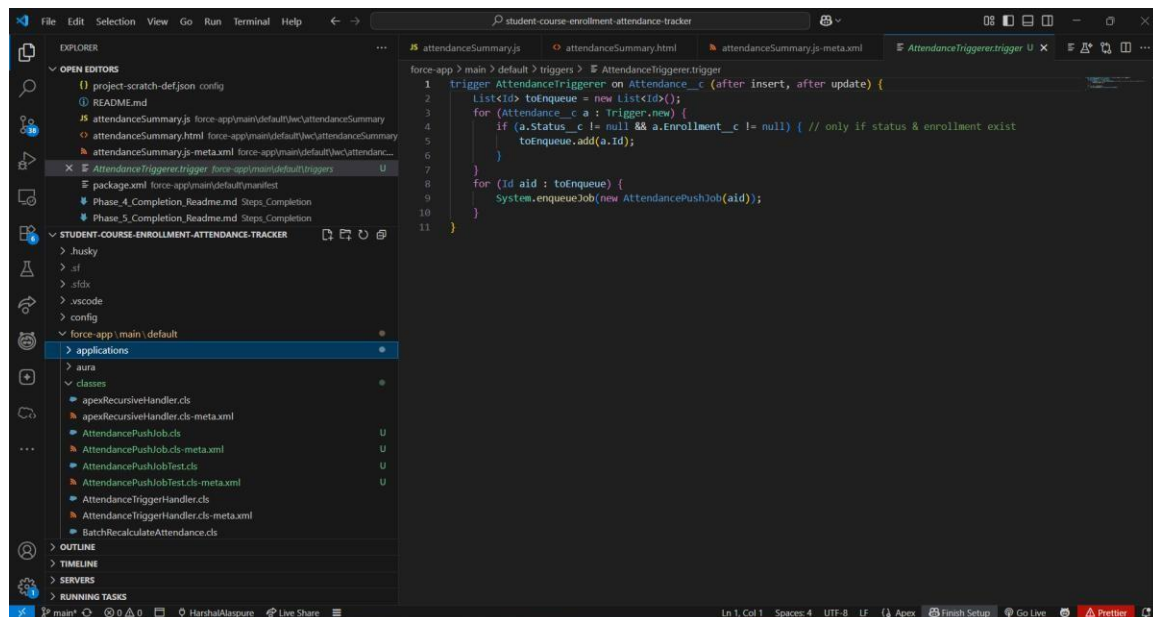
Sort & FilterFind & SelectAdd-insCreate a PDFAdobe Acrobat

Record ID

Record ID	Created By	Created On	Deleted	Email	First Name	Guardian	Guardian C	Guardian F	Last Active	Last Modified	Last Modified	Last Name	Last Ref	Last View	Owner ID	Phone	Student ID	Student Ni	System Mx	Total Atte	Owner ID	Owner Act
a02gK00000005gK0000	09/18/2022	FALSE	developer@harshal	a06gK0000	8.01E+09	Ravindra Alaspure	005gK0000	09/21/2022	Alas	09/23/2022	09/23/2022	005gK0000	7.02E+09	123	Harshal	09/21/2022	40	005gK0000	TRUE	all		
a02gK00000005gK0000	09/18/2022	FALSE	shrutimisa@shruti	a06gK0000	8.01E+09	Raju Misal	005gK0000	09/21/2022	Misal	09/23/2022	09/23/2022	005gK0000	8.79E+09	124	Shruti	09/21/2022	0	005gK0000	TRUE	all		

VS Code & SFDX

Install VS Code → Install Salesforce Extension Pack → Connect to Org (sfdx force:auth:web:login) → Push/Deploy metadata.



Phase 9: Reporting, Dashboards & Security Review

Reports

Reports

Recent

5 Items

Search recent reports...

New Report New Folder

REPORTS	Report Name	Description	Folder	Created By	Created On	Subscribed
Recent	Sample Flow Report: Screen Flows	Which flows run, what's the status of each interview, and how long do users take to complete the screens?	Public Reports	Automated Process	7/17/2025, 9:17 AM	
Created by Me	Grades Reports		Private Reports	Harshal Alaspure	9/26/2025, 12:52 AM	
Private Reports	First_Attendance_Record		Private Reports	Harshal Alaspure	9/26/2025, 12:37 AM	
Public Reports	Enrolled Students Report		Private Reports	Harshal Alaspure	9/26/2025, 12:49 AM	
All Reports	Attendance Records of students	Sorted by date in ascending here are the attendance reports	Private Reports	Harshal Alaspure	9/26/2025, 12:43 AM	

FOLDERS

All Folders

Created by Me

Shared with Me

FAVORITES

All Favorites

Reports → New Report → Object: Enrollment_c → Report: Enrolled Students Report

Report: Students with Enrollments and Courses

Enrolled Students Report

Enable Field Editing

Search

Add Chart

Filter

Copy

Edit

Total Records: 3

Average Total Attendance: 66.67

Student: Student Name	Last Name	Enrollment: Enrollment Number	Course: Course Name	Email	Guardian Name	Total Attendance	Status	
1	Harshal	Alaspure	ENR-0001	Data Structures	[REDACTED]	Ravindra Alaspure	33.33	Enrolled
2	Harshal	Alaspure	ENR-0002	Machine Learning	[REDACTED]	Ravindra Alaspure	33.33	Enrolled
3	Shruti	Misal	ENR-0003	Machine Learning	[REDACTED]	Raju Misal	100.00	Enrolled
4						Avg: 66.67		

Reports → New Report → Object: Attendance_c → Report: Attendance Records of students

Report: Attendance Records with Enrollment

Attendance Records of students

Sorted by date in ascending here are the attendance reports

Enable Field Editing

Search

Add Chart

Filter

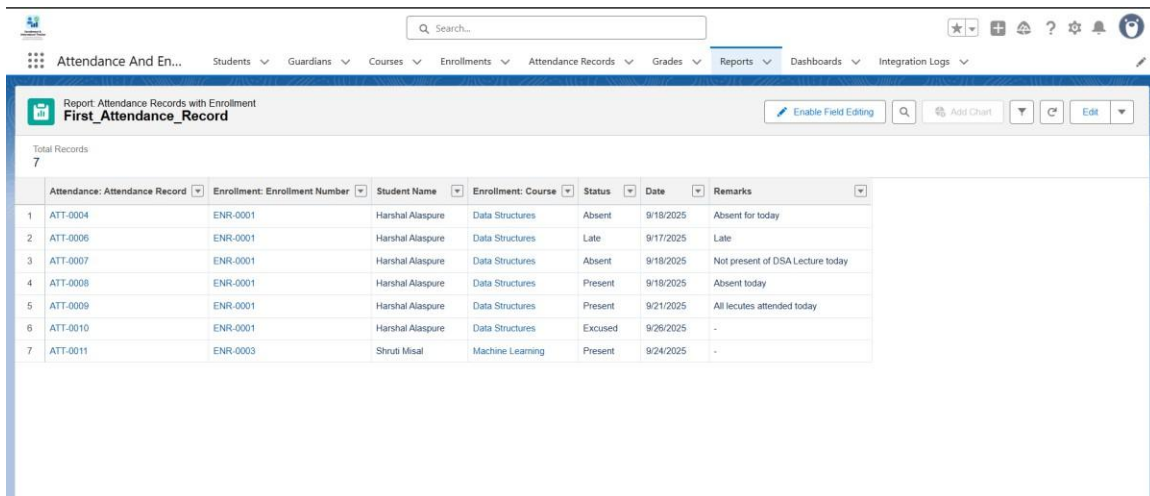
Copy

Edit

Total Records: 7

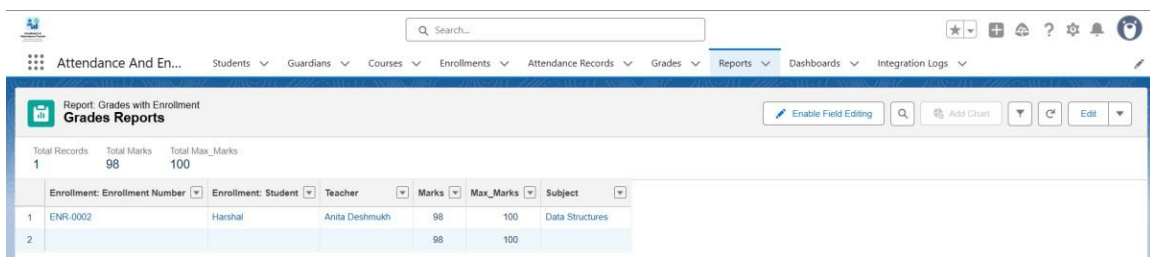
Date	Student Name	Enrollment: Course	Status	Remarks	
1	9/17/2025	Harshal Alaspure	Data Structures	Late	Late
2	9/18/2025	Harshal Alaspure	Data Structures	Absent	Absent for today
3	9/18/2025	Harshal Alaspure	Data Structures	Absent	Not present of DSA Lecture today
4	9/18/2025	Harshal Alaspure	Data Structures	Present	Absent today
5	9/21/2025	Harshal Alaspure	Data Structures	Present	All lectures attended today
6	9/24/2025	Shruti Misal	Machine Learning	Present	-
7	9/26/2025	Harshal Alaspure	Data Structures	Excused	-

Reports → New Report → Object: Attendance_c → Report: First_Attendance_Record



	Attendance: Attendance Record	Enrollment: Enrollment Number	Student Name	Enrollment: Course	Status	Date	Remarks
1	ATT-0004	ENR-0001	Harshal Alaspure	Data Structures	Absent	9/18/2025	Absent for today
2	ATT-0006	ENR-0001	Harshal Alaspure	Data Structures	Late	9/17/2025	Late
3	ATT-0007	ENR-0001	Harshal Alaspure	Data Structures	Absent	9/18/2025	Not present of DSA Lecture today
4	ATT-0008	ENR-0001	Harshal Alaspure	Data Structures	Present	9/18/2025	Absent today
5	ATT-0009	ENR-0001	Harshal Alaspure	Data Structures	Present	9/21/2025	All lectures attended today
6	ATT-0010	ENR-0001	Harshal Alaspure	Data Structures	Excused	9/25/2025	-
7	ATT-0011	ENR-0003	Shruti Misal	Machine Learning	Present	9/24/2025	-

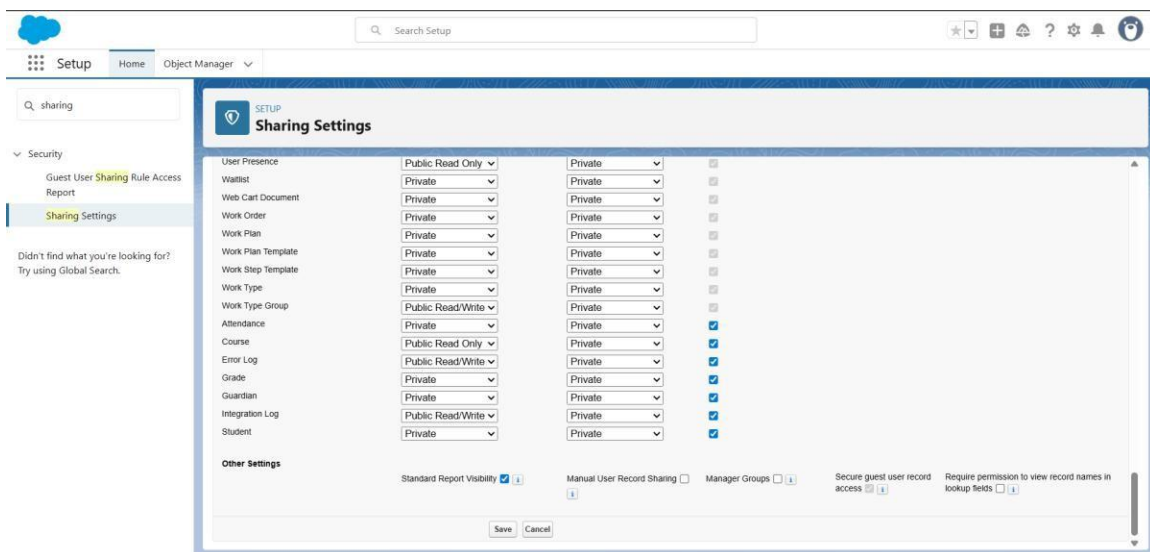
Reports → New Report → Object: Grade_c → Report: Grades Reports



	Enrollment: Enrollment Number	Enrollment: Student	Teacher	Marks	Max_Marks	Subject
1	ENR-0002	Harshal	Anita Deshmukh	98	100	Data Structures
2				98	100	

Sharing Settings (OWD)

Setup → Sharing Settings → Edit OWD → Define object-level access for Attendance, Enrollment, Grade, etc.



Object	Public Read Only	Private	Secure guest user record access	Require permission to view record names in lookup fields
User Presence	☒	☐	☐	☐
Waitlist	☐	☐	☐	☐
Web Cart Document	☐	☐	☐	☐
Work Order	☐	☐	☐	☐
Work Plan	☐	☐	☐	☐
Work Plan Template	☐	☐	☐	☐
Work Step Template	☐	☐	☐	☐
Work Type	☐	☐	☐	☐
Work Type Group	☐	☐	☐	☐
Attendance	☐	☐	☒	☒
Course	☐	☐	☒	☒
Error Log	☐	☐	☐	☐
Grade	☐	☐	☒	☒
Guardian	☐	☐	☐	☐
Integration Log	☐	☐	☒	☒
Student	☐	☐	☒	☒

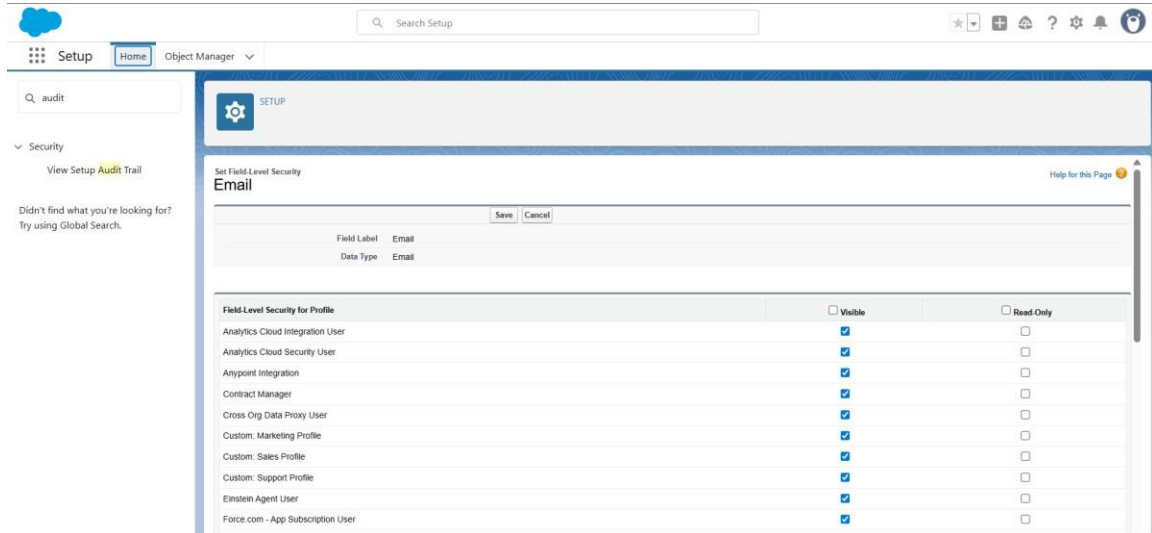
Other Settings

Standard Report Visibility ☒ Manual User Record Sharing ☐ Manager Groups ☐ Secure guest user record access ☐ Require permission to view record names in lookup fields ☐

Save Cancel

Field Level Security

Setup → Object Manager → [Selecting Object] → Fields & Relationships → Select Field → Set Field-Level Security.

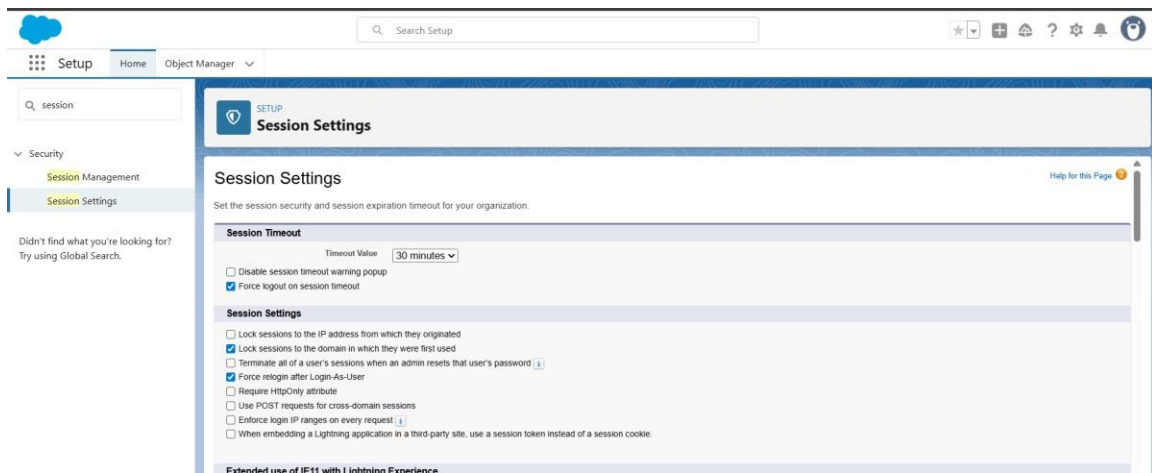


The screenshot shows the Salesforce Setup interface for setting field-level security for the 'Email' field. The left sidebar includes a search bar with 'audit' and a 'Security' section with 'View Setup Audit Trail'. The main content area is titled 'Set Field Level Security Email' and includes 'Save' and 'Cancel' buttons. Below this is a table for 'Field Level Security for Profile'.

Field Label	Email	
Data Type	Email	
Field Level Security for Profile		
		☐ Visible ☐ Read Only
Analytics Cloud Integration User		☒ ☐
Analytics Cloud Security User		☒ ☐
Anypoint Integration		☒ ☐
Contract Manager		☒ ☐
Cross Org Data Proxy User		☒ ☐
Custom: Marketing Profile		☒ ☐
Custom: Sales Profile		☒ ☐
Custom: Support Profile		☒ ☐
Einstein Agent User		☒ ☐
Force.com - App Subscription User		☒ ☐

Session Settings

Setup → Session Settings → Session Timeout = 30 minutes → Save.



The screenshot shows the Salesforce Setup interface for Session Settings. The left sidebar includes a search bar with 'session' and a 'Security' section with 'Session Management' and 'Session Settings'. The main content area is titled 'Session Settings' and includes a description: 'Set the session security and session expiration timeout for your organization.'.

Session Timeout

Timeout Value: 30 minutes

☐ Disable session timeout warning popup

☒ Force logout on session timeout

Session Settings

☐ Lock sessions to the IP address from which they originated

☒ Lock sessions to the domain in which they were first used

☐ Terminate all of a user's sessions when an admin resets that user's password

☒ Force relogin after Login-As-User

☐ Require HttpOnly attribute

☐ Use POST requests for cross-domain sessions

☐ Enforce login IP ranges on every request

☐ When embedding a Lightning application in a third-party site, use a session token instead of a session cookie.

Extended use of IE11 with Lightning Experience

Login IP Ranges

Setup → Profiles → [Selecting Profile] → Login IP Ranges → No restrictions added.

Audit Trail

Setup → View Setup Audit Trail → Download CSV if required.

Setup

Home

Object Manager

Q audit

Security

View Setup Audit Trail

Didn't find what you're looking for?

Try using Global Search.

VIEW

View Setup Audit Trail

Date	User	Source Namespace Prefix	Action	Section	Delegate User
9/26/2025, 1:23:21 AM PDT	alasqureth541@agentforce.com		Changed Session Timeout Value from 120 to 30 minutes	Session Settings	
9/26/2025, 1:23:21 AM PDT	alasqureth541@agentforce.com		Session Security Level for Multi-Factor Authentication was set to High Assurance	Session Settings	
9/26/2025, 1:23:21 AM PDT	alasqureth541@agentforce.com		Session Security Level for Passwordless Login was set to Standard	Session Settings	
9/26/2025, 1:23:21 AM PDT	alasqureth541@agentforce.com		Session Security Level for Lightning Login was set to Standard	Session Settings	
9/26/2025, 1:23:21 AM PDT	alasqureth541@agentforce.com		Session Security Level for Activation was set to Standard	Session Settings	
9/26/2025, 1:23:21 AM PDT	alasqureth541@agentforce.com		Session Security Level for Delegated Authentication was set to Standard	Session Settings	
9/26/2025, 1:23:21 AM PDT	alasqureth541@agentforce.com		Session Security Level for Username Password was set to Standard	Session Settings	
9/25/2025, 10:35:59 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:33:32 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:29:43 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:28:23 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:20:21 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:16:42 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:16:29 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:15:17 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:11:14 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:08:28 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:03:42 PM PDT	alasqureth541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:02:03 PM PDT	alasqureth541@agentforce.com		Created AttendancePushJobTest Apex Class code	Apex Class	

Download setup.audit.trail for last six months (Excel .csv file)

Phase 10: Quality Assurance Testing

Object Record Creation Testing (Student_c)

The screenshot shows a web application interface with a modal form titled "Information" for creating a new student record. The form contains the following fields:

- * Student Name: Rohit
- * Student ID: 125
- * First Name: Rohit
- * Last Name: Sharma
- Email: dummy@dummymail.com
- Phone: 9999999999
- Guardian Name: Gurunaath Sharma
- * Guardian Contact: 9090909090

The form also displays the "Owner" as Harshal Alaspure. At the bottom, there are "Cancel", "Save & New", and "Save" buttons. The background shows a "Students" list with "Recently Viewed" items: Harshal and Shruti.

The screenshot shows the "Student Rohit" details page. A green notification banner at the top states: "Student 'Rohit' was created." The page is divided into two main sections: "Details" and "Activity".

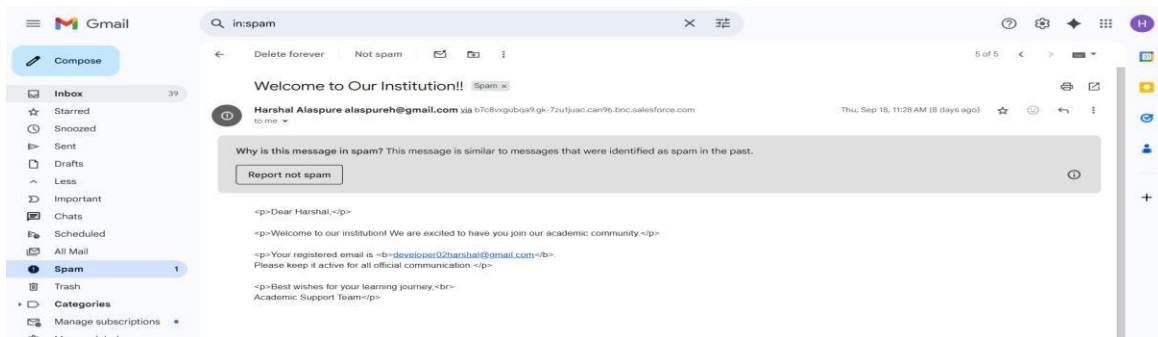
Details Section:

- Student Name: Rohit
- Student ID: 125
- First Name: Rohit
- Last Name: Sharma
- Email: dummy@dummymail.com
- Phone: (999) 999-9999
- Guardian Name: Gurunaath Sharma
- Guardian Contact: (909) 090-9090
- Guardian: Gurunaath
- Total Attendance: 0

Activity Section:

- Filters: All time • All activities • All types
- Buttons: Refresh • Expand All • View All
- Section: Upcoming & Overdue
- Message: No activities to show. Get started by sending an email, scheduling a task, and more.
- Message: No past activity. Past meetings and tasks marked as done show up here.

Welcome mail sent to user



Total % Attendance Generator Trigger on Updating Attendance Record

----Attendance is recorded (As present)

The screenshot shows a 'New Attendance' modal form. The form is titled 'New Attendance' and has a close button (X) in the top right corner. It contains several fields: 'Date' (9/26/2025), 'Status' (Present), 'Marked By' (Anita Deshmukh), 'Remarks' (Present today), and 'Enrollment' (ENR-0001). The form also shows the 'Owner' as Harshal Alaspure. At the bottom, there are buttons for 'Cancel', 'Save & New', and 'Save'. The background shows a sidebar with 'Attendance And En...' and 'Students' tabs, and a main area with 'Attendance Records' and 'Recently Viewed' sections.

The screenshot shows the 'Attendance' record for Harshal Alaspure. The record is displayed in a table with columns: Date, Status, Enrollment, and Marked By. The data row shows: Date: 9/26/2025, Status: Present, Enrollment: ENR-0001, and Marked By: Anita Deshmukh. A green notification banner at the top says 'Attendance "ATT-0012" was created.' The background shows a sidebar with 'Attendance And En...' and 'Students' tabs, and a main area with 'Attendance Records' and 'Recently Viewed' sections.

So once it is created % attendance is triggered in Respective student object (Harshal here)

The screenshot displays a CRM interface with a top navigation bar containing tabs like 'Attendance And En...', 'Students', 'Guardians', 'Courses', 'Enrollments', 'Attendance Records', 'Grades', 'Reports', 'Dashboards', and 'Integration Logs'. The main section shows the 'Student' profile for 'Harshal'. The profile details include:

- Student Name: Harshal
- Student ID: 123
- First Name: Harshal
- Last Name: Alaspure
- Email: [Redacted]
- Phone: [Redacted]
- Guardian Name: Ravindra Alaspure
- Guardian Contact: [Redacted]
- Guardian: Ravindra
- Total Attendance: 42.86

Additional information includes 'Created By: Harshal Alaspure, 9/17/2025, 10:57 PM' and 'Last Modified By: Harshal Alaspure, 9/26/2025, 4:16 AM'. On the right, there are buttons for 'New Contact', 'Edit', and 'New Opportunity', along with filters and a section titled 'Upcoming & Overdue'.

And an automated email is sent to respective student with status listed

The screenshot shows a Gmail inbox with the search filter 'in:spam'. The email in question is titled 'Attendance Update for 2025-09-26' and is marked as 'Spam'. The sender is 'Harshal Alaspure alaspureh@gmail.com'. The email content is as follows:

Hello Harshal,

Your attendance for September 26, 2025 has been recorded as: Excused

Remarks:

Best regards,
Academic Office

The email interface also shows a sidebar with folders like 'Compose', 'Inbox', 'Starred', 'Snooted', 'Sent', 'Drafts', 'Less', 'Important', 'Chats', 'Scheduled', 'All Mail', 'Spam', 'Trash', and 'Categories'.